

Notes of Optimization

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LECTURE 1 OF 03/03/2026

We will give a quick summary on basic notions of topology, calculus 1 & 2.

DEFINITION 1 (Norm/scalar product). Given $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ the Euclidean norm (or ℓ^2 norm) is defined as

$$\|x\| = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}.$$

For $x, y \in \mathbb{R}^n$ the *scalar product* is given by

$$\langle x, y \rangle = x \cdot y = x^\top y = \sum_{i=1}^n x_i y_i.$$

Moreover, the scalar product induces a norm. Indeed, one can verify that

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

DEFINITION 2 (Open ball). Given $x \in \mathbb{R}^n$, $r > 0$ we call the *open ball of radius r centered in x* the set

$$B(x, r) = \{y \in \mathbb{R}^n : \|y - x\| < r\}.$$

DEFINITION 3 (Limit of a sequence). Let $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}^n$, $\bar{x} \in \mathbb{R}^n$. We say that x_n converges to $\bar{x}$, and write $x_n \rightarrow \bar{x}$ if for every $\varepsilon > 0$ there exists an $N > 0$ such that for every $k > N$, $x_k \in B(\bar{x}, \varepsilon)$, i.e. for k big enough, $\|\bar{x} - x_k\| < \varepsilon$, and we say that

$$\lim_{k \rightarrow +\infty} x_k = \bar{x}.$$

Remark 4. The limit in $\mathbb{R}^n$ is the pointwise limit, so $x_k \rightarrow \bar{x}$ if and only if $(x_k)_j \rightarrow (\bar{x})_j$ for any $1 \leq j \leq n$.

Example 5. Consider the sequence $\{x_k\}_{k \in \mathbb{N}}$ defined as

$$x_k = \left(\frac{1}{k}, \frac{1}{k} \sin(k) \right) \rightarrow (0, 0)$$

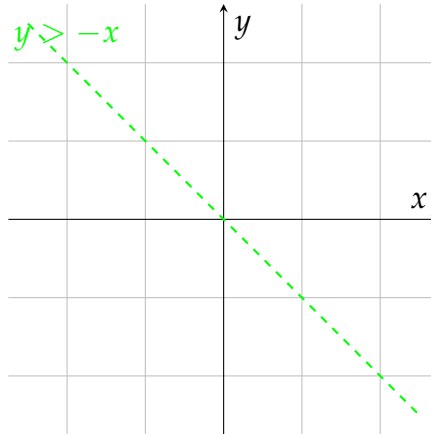


Figure 1.1: Ball of radius r centered in x on $\mathbb{R}^2$.

as $k \rightarrow +\infty$.

DEFINITION 6 (Open/closed set). Let $A \subseteq \mathbb{R}^n$. We say that A is *open* if for every $x \in A$ there exists an $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$. We say that A is *closed* if $\mathbb{R}^n \setminus A$ is open.

Example 7. Consider the set

$$A = \left\{ (x_1, x_2) \in \mathbb{R}^2 : |x_1| < 1, |x_2| < 1 \right\}.$$

We can easily verify that A is open. Observe that a set can be neither open nor closed. The sets $\mathbb{R}^n$ and $\emptyset$ are both open and closed at the same time.

DEFINITION 8 (Neighbour). We say that, for a given $x \in \mathbb{R}^n$, a subset $A \subseteq \mathbb{R}^n$ is a *neighbour* of x if A is an open set containing x .

DEFINITION 9 (Interior, exterior, boundary, closure). Let $A \subseteq \mathbb{R}^n$, $x \in \mathbb{R}^n$. We say that x is:

- an *interior point* for A if $\exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$. We denote the set of all interior points of A with A° ;
- A *boundary point* for A if $\forall \varepsilon > 0$, $B(x, \varepsilon) \cap A \neq \emptyset$ and $B(x, \varepsilon) \cap A^c \neq \emptyset$. We denote the set of all the boundary points with ∂A ;
- an *exterior point* for A if $\exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A^c$.

We denote the *closure* of A as $\bar{A} = A \cup \partial A$.

PROPOSITION 10. Let $\{A_k\}_{k \in \mathbb{N}}$ a sequence of open sets. Then:

i) the countable union of open sets is open, i.e.

$$\bigcup_{k \in \mathbb{N}} A_k \text{ is open;}$$

ii) the finite intersection of open sets is open, i.e. for $N \in \mathbb{N}$,

$$\bigcap_{i=1}^N A_i \text{ is open.}$$

Proof. We prove i) and ii) separately.

i): set $E = \bigcup_{k=1}^{+\infty} A_k$. Given $x \in E$ then there exists $m > 0$ such that $x \in A_m$. Since A_m is open, there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A_m \subseteq E$, so we found an open ball containing x fully contained in E . Since the choice of x is arbitrary, this must hold for every $x \in E$, hence E is open.

ii): set $E = \bigcap_{k=1}^N A_k$. Let $x \in E$, then $x \in A_k$ for every $1 \leq k \leq N$, hence there exists a sequence $\varepsilon_k > 0$ such that $B(x, \varepsilon_k) \subseteq A_k$ for every $k < N$. Consider $\bar{\varepsilon} = \frac{1}{2} \min_k \varepsilon_k > 0$. Then $B(x, \bar{\varepsilon}) \subseteq B(x, \varepsilon_k)$ for every k , hence $B(x, \bar{\varepsilon}) \subseteq E$. Again, since the choice of x is arbitrary, this must hold true for every $x \in E$, hence E is open. ■

PROPOSITION 11 (Characterization of closed sets). Let $C \subseteq \mathbb{R}^n$. Then C is closed $\iff$ for every sequence $\{x_k\}_{k \in \mathbb{N}} \subseteq C$ such that $x_k \rightarrow \bar{x}$, then $\bar{x} \in C$.

Proof. We prove $\implies$ and $\impliedby$.

" $\implies$ ": let $C \subseteq \mathbb{R}^n$ be closed. Consider $\{x_k\}_{k \in \mathbb{N}}$ such that $x_k \rightarrow \bar{x} \in \mathbb{R}^n$, we want to show that $\bar{x} \in C$. Assume, by contradiction, that $\bar{x} \notin C \implies \bar{x} \in \mathbb{R}^n \setminus C$, which is an open set, so there exists $\varepsilon > 0$ such that $B(\bar{x}, \varepsilon) \subseteq \mathbb{R}^n \setminus C \implies B(\bar{x}, \varepsilon) \cap C = \emptyset$, but $x_k \rightarrow \bar{x}$, so, for this ε , there must be an $N > 0$ such that $x_k \in B(\bar{x}, \varepsilon)$ for $k > N$, which is in contradiction with the fact that the sequence takes values in C .

" $\Leftarrow$ ": let C be a set. Consider $\{x_k\}_{k \in \mathbb{N}} \subseteq C$ such that $x_k \rightarrow \bar{x} \in C$. Assume, by contradiction, C not closed. Then, for every $\varepsilon > 0$, there exists $z \in \mathbb{R}^n \setminus C$ such that $B(z, \varepsilon) \cap (\mathbb{R}^n \setminus C) = \emptyset$ and $B(z, \varepsilon) \cap C \neq \emptyset$. So we can find a sequence $\{x_k\}_{k \in \mathbb{N}}$ with values in C such that $x_k \rightarrow z$ by fixing a sequence $\{\varepsilon_k\}_{k \in \mathbb{N}}$ such that $\varepsilon_k \searrow 0$, so $x_k \in B(z, \varepsilon_k) \cap C$, $x_k \rightarrow z$, so, by assumption $z \in C$, which is in contradiction with $z \in \mathbb{R}^n \setminus C$. ■

DEFINITION 12 (Bounded/finite set). A set $A \subseteq \mathbb{R}^n$ is *bounded* if there exists $R > 0$ such that $A \subseteq B(0, R)$.

The set A is *finite* if $A = \{x_1, \dots, x_m\} \subseteq \mathbb{R}^n$.

DEFINITION 13 (Continuity). Let $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\bar{x} \in E$. We say that f is *continuous* in $\bar{x}$ if for every sequence $\{x_k\}_{k \in \mathbb{N}}$ such that $x_k \rightarrow \bar{x}$ in $\mathbb{R}^n$, then $f(x_k) \rightarrow f(\bar{x})$ in $\mathbb{R}^m$.

DEFINITION 14 (Cluster/isolated point). Let $E \subseteq \mathbb{R}^n$, $\bar{x} \in \mathbb{R}^n$. We say that $\bar{x}$ is a *cluster point* for E if there exists a sequence $\{x_k\}_{k \in \mathbb{N}}$ such that, for every $\bar{x} \neq x_k \rightarrow \bar{x}$. We denote the set of all cluster points with $\text{Acc}(E)$.

A point $\bar{x}$ is *isolated* in E if $x \in E \setminus \text{Acc}(E)$.

DEFINITION 15 (Limit). Let $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\bar{x} \in \text{Acc}(E)$. Fixed $\ell \in \mathbb{R}^m$, we say that

$$\lim_{x \rightarrow \bar{x}} f(x) = \ell$$

if for every sequence $\{x_k\}_{k \in \mathbb{N}}$, $x_k \rightarrow \bar{x}$, $x_k \neq \bar{x}$ for every k , we have $f(x_k) \rightarrow \ell$. Notice that we allow $\bar{x}$ to not necessarily be in E .

Example 16. Consider the function $f : \{0\} \cup [1, 2] \subset \mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x$. Then it has no meaning to speak about $\lim_{x \rightarrow 0} f(x)$, as $0 \notin \text{Acc}(\{0\} \cup [1, 2])$, while f is continuous at 0, since the only sequence converging to 0 in $\{0\} \cup [1, 2]$ is the sequence which is identically 0.

Remark 17. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous, then:

- $\{x : f(x) < c\}$ is *open* $\forall c \in \mathbb{R}$;
- $\{x : f(x) \leq c\}$ is *closed* $\forall c \in \mathbb{R}$;
- $\{x : f(x) = c\}$ is *closed* $\forall c \in \mathbb{R}$.

CHAPTER 2

LECTURE 2 OF 04/03/2026

Our main focus from now on is to solve the problem

$$\inf_{x \in A} f(x)$$

for some $A \subseteq \mathbb{R}^n$ and some $f : \mathbb{R}^n \rightarrow \mathbb{R}$. We will restrict ourselves only to certain functions and sets.

DEFINITION 1 (Infimum/supremum). Let $E \subseteq \mathbb{R}$. We denote with $\inf E =$ largest $z \in \mathbb{R}$ such that $z \leq x$ for every $x \in E$. By convention we set $\inf \emptyset = +\infty$.

Similarly, we denote with $\sup E$ the smallest $z \in \mathbb{R}$ such that $z \geq x$ for every $x \in E$, with the convention $\sup \emptyset = -\infty$. More formally,

$$\begin{aligned} z = \inf E & \quad \text{if } z \leq x \forall x \in E \text{ and } \forall \varepsilon > 0 \exists y \in E \text{ s.t. } y \leq z + \varepsilon \\ z = \sup E & \quad \text{if } z \geq x \forall x \in E \text{ and } \forall \varepsilon > 0 \exists y \in E \text{ s.t. } y \geq z - \varepsilon \end{aligned}$$

Remark 2. The inf and sup aren't necessarily inside the set. If they are, we speak of min and max.

Axiom 1 — Completeness of $\mathbb{R}$

Every bounded sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}$ admits a supremum in $\mathbb{R}$.

THEOREM 3 (Bolzano-Weierstrass). Any bounded sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}$ admits a cluster point, i.e. there exists $\bar{x} \in \mathbb{R}$ and a subsequence $\{x_{n_j}\}_{j \in \mathbb{N}}$ such that $x_{n_j} \rightarrow \bar{x}$ as $j \rightarrow +\infty$.

Proof. Since $\{x_n\}_n$ is bounded, there exists $R > 0$ such that $x_n \in [-R, R]$ for every $n \in \mathbb{N}$. Set $I_1 = [-R, R]$, select x_{n_1} a value in I_1 . Divide I_1 in half and select the one subinterval containing infinitely many x_n s, call it $I_2 = [a_2, b_2]$ (here $a_2 = -R, b_2 = 0$ or $a_2 = 0, b_2 = R$). We have $I_2 \subset I_1$, $|I_2| = |I_1|/2$, and choose x_{n_2} a value in I_2 . Repeat this process to cut and select. We end up, at the k -th step, with a sequence of subintervals $I_1, I_2, \dots, I_k$ and 3 subsequences: $x_{n_1}, x_{n_2}, \dots, x_{n_k}$

the sequence of selected values, $a_1, a_2, \dots, a_k$ the sequence of left boundaries and $b_1, b_2, \dots, b_k$ the sequence of right boundaries.

We observe that $a_{j+1} \geq a_j$ and $b_{j+1} \leq b_j$ for all $1 \leq j \leq k$. Moreover, $a_j \leq R$ and $b_j \geq -R$ for all j , so, by the axiom of completeness of $\mathbb{R}$,

- there exists $\bar{a} \in \mathbb{R}$ such that $a_k \rightarrow \bar{a} = \sup_k a_k$
- there exists $\bar{b} \in \mathbb{R}$ such that $b_k \rightarrow \bar{b} = \inf_k b_k$

We observe that $\bar{a} = \bar{b}$. Indeed, $|\bar{b} - \bar{a}| = |\bar{b} - b_k + b_k - a_k + a_k - \bar{a}| \leq |\bar{b} - b_k| + |\bar{a} - a_k| + |b_k - a_k|$. The terms $|\bar{b} - b_k|$ and $|\bar{a} - a_k|$ goes to 0 as $k \rightarrow +\infty$ for what we observed before. Moreover, $|b_k - a_k| = 2R/2^{k-1} \rightarrow 0$ as $k \rightarrow +\infty$.

Set $\bar{x} = \bar{a} (= \bar{b})$. Then the subsequence $\{x_{n_j}\}_j$ is converging to $\bar{x}$: indeed $|x_{n_j} - \bar{x}| = |x_{n_j} - a_j + a_j - \bar{x}| \leq |a_{n_j} - a_j| + |a_j - \bar{x}|$.

Since $a_j \rightarrow \bar{x} = \bar{a}$, the term $|a_j - \bar{x}| \rightarrow 0$. Notice that, by construction, $x_{n_j} \in I_j$, so $|x_{n_j} - a_j| \leq |I_j| \rightarrow 0$.

■

DEFINITION 4. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. We say that $\bar{x} \in \mathbb{R}^n$ is a *local maximum* for f if there exists a neighbourhood U of $\bar{x}$ such that $f(x) \leq f(\bar{x})$ for every $x \in U$. A *local strict maximum* has the strict inequality.

Similarly, $\bar{x}$ is *local minimum* for f if $\exists U$ neighbourhood of $\bar{x}$ such that $f(x) \geq f(\bar{x})$ for all $x \in U$.

If there exists $\bar{x}$ such that $f(x) = \inf_{x \in A} f(x)$, the $\bar{x}$ is a *global minimum* over A .

DEFINITION 5. We denote with $\arg_{x \in A} \min f(x)$ the set of minimum points for f over A , i.e

$$\arg_{x \in A} \min f(x) = \left\{ \bar{x} \in \mathbb{R}^n : f(\bar{x}) = \inf_{x \in A} f(x) \right\}.$$

THEOREM 6 (Weierstrass). Let $A \subseteq \mathbb{R}^n$ be a compact set, $f : A \rightarrow \mathbb{R}$ a continuous function. Then there exists $x_m, x_M \in A$ such that

$$f(x_m) = \min_{x \in A} f(x)$$

$$f(x_M) = \max_{x \in A} f(x).$$

Proof. We prove the statement for the minimum point.

Let $\ell = \inf_{x \in A} f(x)$. We can always find a sequence $\{x_n\}_n \subseteq A$ such that $f(x_n) \rightarrow \ell$ by the continuity of f . Since A is bounded, $\{x_n\}_n$ is bounded as well. By Bolzano-Weierstrass, there exists a subsequence $\{x_{n_k}\}_k$ converging to $\bar{x} \in A$ by closeness of A . By continuity, in particular f is continuous at $\bar{x}$, ■

Remark 7. For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we define the *gradient* of f as

$$\nabla f(c) = \left(\frac{\partial f}{\partial x_1}(c), \dots, \frac{\partial f}{\partial x_n}(c) \right)$$

the vector of partial derivatives, defined as

$$\frac{\partial f}{\partial x_i}(c) = \lim_{h \rightarrow 0} \frac{f(c + he_i) - f(c)}{h}$$

whenever this limit exists finite, and where e_i is the i -th vector of the canonical base of $\mathbb{R}^n$. Furthermore, we say that f is *differentiable* at c if

$$f(x) = f(c) + \langle \nabla f(c), x - c \rangle + \mathcal{O}(\|x - c\|^2)$$

for $x \rightarrow c$.

We now set our focus on convex analysis.

DEFINITION 8 (Convex set). Let $C \subseteq \mathbb{R}^n$. We say that C is *convex* if, for any $x, y \in C$, and for any $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha)y \in C$.

PROPOSITION 9 (Book 1.1.1). a) $\bigcap_{i \in I} C_i$ is convex if C_i is convex for any $i \in I$ finite or countable;

b) the algebraic sum $C_1 + C_2 = \{x + y : x \in C_1, y \in C_2\}$ is convex if C_1 and C_2 are convex;

c) the set $\lambda C = \{\lambda x : x \in C\}$ is convex if C is convex for any $\lambda \in \mathbb{R}$;

d) $\overline{C}, C^\circ$ are convex if C is convex;

e) if a function $f : C \rightarrow \mathbb{R}$ is affine, then the set $f(C) = \{y : y = f(x), x \in C\}$ is convex if C is convex.

Proof. We prove all the propositions separately.

a): suppose C_i convex and let $C = \bigcap_{i \in I} C_i$. Let $x, y \in C, \alpha \in [0, 1]$. Since $x, y \in C$, in particular $x, y \in C_i$ for all $i \in I$. But then also $\alpha x + (1 - \alpha)y \in C_i$ for all $i \in I$, hence C is convex.

b): let $x = x_1 + x_2 \in C_1 + C_2, x_1 \in C_1$ and $x_2 \in C_2$, take $y \in C_1 + C_2$ of the same form.

$$\alpha x + (1 - \alpha)y = \underbrace{\alpha x_1 + (1 - \alpha)y_1}_{\in C_1} + \underbrace{\alpha x_2 + (1 - \alpha)y_2}_{\in C_2} \in C.$$

c): trivial.

d): consider $\{x_k\}_k \subseteq C$ and $\{y_k\}_k \subseteq C$ such that $x_k \rightarrow x$ and $y_k \rightarrow y$. Then the sequence $\{\alpha x_k + (1 - \alpha)y_k\}_k \subseteq C$. But then also $\alpha x + (1 - \alpha)y \in C$ as limit of sequences valued in C .

Let now $x, y \in C^\circ$. Then there exists $r > 0$ such that $B(x, r) \subseteq C$ and $B(y, r) \subseteq C$. Consider $\alpha x + (1 - \alpha)y$. Our claim is: $B(\alpha x + (1 - \alpha)y, r) \subseteq C$. Indeed, any $z \in B(\alpha x + (1 - \alpha)y, r)$ is of the form $z = \alpha x + (1 - \alpha)y + w$ with $\|w\| < r$, so $\alpha(x + w) + (1 - \alpha)(y + w) \in C$ by convexity of C , hence the claim. ■

LECTURE 3 OF 10/03/2026

Example 1 (Convex sets). The following sets are convex in $\mathbb{R}^n$:

- **hyperplanes**: $\{x \in \mathbb{R}^n : a^\top x = b\}$ for $a \in \mathbb{R}^n, b \in \mathbb{R}$;
- **halfspaces**: $\{x \in \mathbb{R}^n : a^\top x \leq b\}$ for $a \in \mathbb{R}^n, b \in \mathbb{R}$;
- **polyhedra**: $\{x \in \mathbb{R}^n : a_i^\top x \leq b_i\}$ for $a_i \in \mathbb{R}^n, b_i \in \mathbb{R}$ for $i = 1, \dots, m$ (they are intersection of half spaces).

Half cones are convex as well.

DEFINITION 2 (Convex/concave function). Let $C \subseteq \mathbb{R}^n$ be convex. Consider $f : C \rightarrow \mathbb{R}$. We say that f is *convex* if the following holds:

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y) \quad \forall x, y \in C, \forall \alpha \in [0, 1].$$

We say that f is *concave* if $-f$ is convex.

Example 3. Affine functions are convex. Indeed, let $f(x) = a^\top x + b, a \in \mathbb{R}^n$.
Indeed,

$$\begin{aligned} f(\alpha x + (1 - \alpha)y) &= \langle a, \alpha x + (1 - \alpha)y \rangle + b(1 - \alpha + \alpha) \\ &= \alpha \langle a, x \rangle + \alpha b + (1 - \alpha) \langle a, y \rangle + (1 - \alpha)b \\ &= \alpha f(x) + (1 - \alpha)f(y). \end{aligned}$$

Example 4. The norm in $\mathbb{R}^n$ is convex, as an easy consequence of triangular inequality.

We can observe that f convex implies that the set $V_\gamma = \{x \in \mathbb{R}^n : f(x) \leq \gamma\}$ is convex for all $\gamma \in \mathbb{R}$. Indeed, for every $x, y \in V_\gamma$ we have $f(x) \leq \gamma$ and $f(y) \leq \gamma$. We see that $f(\alpha x + (1 - \alpha)y) \in V - \gamma$. Indeed, by convexity of f we have $f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y) = \alpha\gamma + (1 - \alpha)\gamma = \gamma$.

In general, the converse is not true, i.e. convex level sets *does not imply* f convex.
From now on, $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$, where $\overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}$.

DEFINITION 5 (Epigraph). Let $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$. We define the set

$$\text{epi}(f) = \left\{ (x, w) \in \mathbb{R}^{n+1} : x \in X, w \in \mathbb{R}, f(x) \leq w \right\}.$$

An intuitive definition could be "the set of points above the graph of f ".

DEFINITION 6 (Effective domain). Given $f : X \subseteq \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ we define the *effective domain* of f as the set

$$\text{dom}(f) = \{x \in X : f(x) < +\infty\} = \{x \in X : \exists w \in \mathbb{R} \text{ s.t. } (x, w) \in \text{epi}(f)\}.$$

DEFINITION 7 (Proper function). A function $f : X \subseteq \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ is said *proper* if $f(x) < +\infty$ and $f(x) > -\infty$ for all $x \in X$. In other words, $\text{epi}(f)$ does not contain any vertical line.

DEFINITION 8 (Extension of convex functions). Let $C \subseteq \mathbb{R}^n$ be a convex set. A function $f : C \rightarrow \overline{\mathbb{R}}$ is *convex* if $\text{epi}(f)$ is convex in $\mathbb{R}^{n+1}$.

Remark 9. If $f : C \rightarrow \mathbb{R}$ the definitions coincides.

Proof. Assume $\text{epi}(f)$ convex in $\mathbb{R}^{n+1}$, then the points $(x, f(x)), (y, f(y)) \in \text{epi}(f)$, but also $\alpha(x, f(x)) + (1 - \alpha)(y, f(y)) \in \text{epi}(f)$. This implies $(\alpha x + (1 - \alpha)y, \alpha f(x) + (1 - \alpha)f(y)) \in \text{epi}(f) \implies f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$. ■

DEFINITION 10 (Convex indicator function). Given $A \subseteq \mathbb{R}^n$, we define the *convex indicator function* of the set A as

$$\delta_A(x) = \begin{cases} 0 & \text{if } x \in A \\ +\infty & \text{otherwise} \end{cases}$$

This is coherent with the infimum operation, as

$$\inf_{x \in A} f(x) = \inf_{x \in \mathbb{R}^n} f(x) + \delta_A(x).$$

DEFINITION 11 (Liminf). Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}, c \in \text{Acc}(X)$. We say that

$$\ell = \liminf_{x \rightarrow c} f(x)$$

if $\ell = \inf_{x_k} \lim_{k \rightarrow +\infty} f(x_k)$ where $\{x_k\}_k$ is a sequence in X , $\{x_k\}_k \rightarrow c$, $x_k \neq c$ for all k and $\lim_{k \rightarrow \infty} f(x_k)$ exists finite.

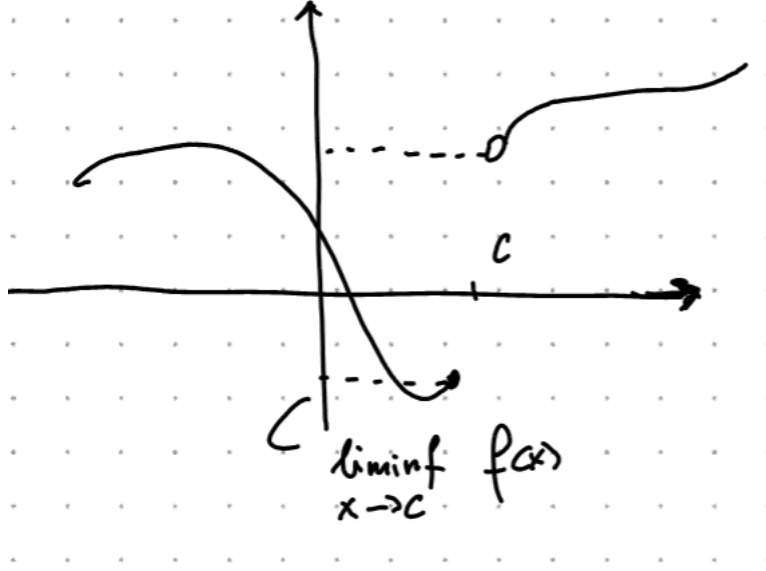


Figure 3.1: The $\liminf$ of f at $x = 1$ is 1, even if $f(1) = 0.5$.

DEFINITION 12 (Lower semicontinuity). Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. We say that f is *lower semicontinuous* (lsc for short) at $c \in X$ if

$$f(c) \leq \liminf_{x \rightarrow c} f(x).$$

THEOREM 13 (Weierstrass). *The Weierstrass theorem still allows a minimum point for a lsc function.*

PROPOSITION 14 (Book 1.1.2). Let $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$. The following are equivalent:

- i) The level sets $V_\gamma = \{x \in \mathbb{R}^n : f(x) \leq \gamma\}$ are closed for all $\gamma \in \mathbb{R}$;
- ii) f is lsc;
- iii) $\text{epi}(f)$ is closed in $\mathbb{R}^{n+1}$.

Proof. We prove $i) \implies ii) \implies iii) \implies i)$.

$i) \implies ii)$: suppose V_γ closed for all γ . Assume by contradiction f not lsc, so there exists $c \in \mathbb{R}^n$ s.t. $f(c) > \liminf_{x \rightarrow c} f(x)$. This implies that there exists a sequence $\{x_k\}_k$, $x_k \rightarrow c$ and a $\gamma \in \mathbb{R}$ s.t. $f(c) > \gamma \geq \lim_{k \rightarrow +\infty} \inf f(x_k)$. For k big enough, we have $f(x_k) \leq \gamma$, so $x_k \in V_\gamma$. By closeness of V_γ , it must be $c \in V_\gamma \implies f(c) \leq \gamma$, which is in contradiction with $f(c) > \gamma$.

ii) $\implies$ iii): assume f lsc, we want to show that any sequence $\{(x_k, w_k)\}_k \subseteq \text{epi}(f)$ converging to $(\bar{x}, \bar{w})$ must satisfy $(\bar{x}, \bar{w}) \in \text{epi}(f)$. Assume $x_k \rightarrow \bar{x}$, $w_k \rightarrow \bar{w}$. Then $f(x_k) \leq w_k$ for all k . Since $x_k \rightarrow \bar{x}$ and f is lsc, it must be that $f(\bar{x}) \leq \liminf_{k \rightarrow +\infty} f(x_k) \leq \liminf_{k \rightarrow +\infty} w_k = \bar{w}$, hence $(\bar{x}, \bar{w}) \in \text{epi}(f)$.

iii $\implies$ i): assume $\text{epi}(f)$ closed. Let $\gamma \in \mathbb{R}$, $V_\gamma = \{x : f(x) \leq \gamma\}$. Consider the sequence $\{x_k\}_k \subseteq V_\gamma$ s.t $x_k \rightarrow \bar{x}$. We want to show $\bar{x} \in V_\gamma$. The points $(x_k, \gamma) \in \text{epi}(f)$ for all k , so, by closeness of $\text{epi}(f)$, $(\bar{x}, \gamma) \in \text{epi}(f)$. This implies $f(\bar{x}) \leq \gamma$, hence $\bar{x} \in V_\gamma$. ■

PROPOSITION 15 (Book 1.1.4). Let $f : \mathbb{R}^m \rightarrow (-\infty, +\infty]$, $A \in \mathbb{R}^{m \times n}$, $F : \mathbb{R}^n \rightarrow (-\infty, +\infty]$, $x \mapsto f(Ax)$. If f is convex, then F is convex. If f is closed, then F is closed.

Proof. Assume f convex. Then

$$F(\alpha x + (1 - \alpha)y) = f(\alpha Ax + (1 - \alpha)Ay) \leq \alpha f(Ax) + (1 - \alpha)f(Ay).$$

Assume f closed, so $\text{epi}(f)$ is closed and f is lsc. Take a sequence $x_k \rightarrow \bar{x}$. Then $\liminf_k F(x_k) = \liminf_k f(Ax_k)$. Let $y_k = Ax_k$, so $y_k \rightarrow A\bar{x}$ by continuity of the linear map Ax . This implies $\liminf_k f(Ax_k) \geq f(A\bar{x}) = F(\bar{x})$. ■

PROPOSITION 16. Let $\{f_i\}_{i=1}^m$ be a family of convex and closed function s.t. $f_i : \mathbb{R}^n \rightarrow (-\infty, +\infty]$, and $\gamma_1, \dots, \gamma_m > 0$. Then the function defined by $F(x) = \sum_{i=1}^m \gamma_i f_i(x)$ is closed and convex.

PROPOSITION 17 (Book 1.1.6). Let $f_i : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ for $i \in I$ any countable index set. Let $f(x) = \sup_i f_i(x)$. If f_i is closed convex for all $i \in I$ then f is closed convex.

Proof. Let $(x, w) \in \text{epi}(f)$, then $f(x) \leq w$, so $f_i(x) \leq w$ for all i . This implies $(x, w) \in \text{epi}(f_i)$ for all i , hence $\text{epi}(f) = \bigcap_{i \in I} \text{epi}(f_i)$, which is convex as countable intersection of convex sets. The same argument holds for closeness. ■

LECTURE 4 OF 11/03/2026

We ask ourselves what happens if f is convex and we add some regularity hypothesis. We first remark that if f is differentiable at c , near this point, the function looks like an hyperplane. The intuition is: function must be always above its tangent hyperplane.

PROPOSITION 1 (Book 1.1.7). *Let $C \subseteq \mathbb{R}^n$ be convex, $A \subseteq \mathbb{R}^n$ open s.t. $C \subseteq A$. Let $f : A \rightarrow \mathbb{R}$ convex and differentiable. Then*

- i) f is convex on $C \iff f(z) \geq f(x) + \langle \nabla f(x), z - x \rangle$ for all $x, z \in C$
- ii) f is strictly convex over $C \iff f(z) > f(x) + \langle \nabla f(x), z - x \rangle$ for all $x, z \in C$ s.t. $x \neq z$

Proof. Let f be convex on C . Let $x, z \in C$. Observe that

$$\lim_{h \rightarrow 0^+} \frac{f(x + h(z - x)) - f(x)}{h} = \frac{\partial}{\partial h} f(x + h(z - x))|_{h=0} = \nabla f(x) \cdot (z - x).$$

Then

$$\begin{aligned} f(x) + \langle \nabla f(x), z - x \rangle &= f(x) + \lim_{h \rightarrow 0^+} \frac{f(x + h(z - x)) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{hf(x) - f(x) + f(hz + (1 - h)x)}{h} \\ &\leq \lim_{h \rightarrow 0} \frac{-(1 - h)f(x) + hf(z) - (1 - h)f(x)}{x} \\ &= \lim_{h \rightarrow 0} f(z) = f(z). \end{aligned}$$

Let now $\alpha \in [0, 1]$, $x, y \in C$, $z = \alpha x + (1 - \alpha)y$.

$$\begin{aligned} f(x) &\geq f(z) + \langle \nabla f(z), x - z \rangle \cdot \alpha \\ f(y) &\geq f(z) + \langle \nabla f(z), y - z \rangle \cdot (1 - \alpha) \end{aligned}$$

Take the sum of the above quantities, then

$$\alpha f(x) + (1 - \alpha)f(y) \geq \alpha f(z) + (1 - \alpha)f(z) + \alpha \langle \nabla f(z), x - z \rangle + (1 - \alpha) \langle \nabla f(z), y - z \rangle$$

hence, by linearity of the scalar product,

$$\alpha f(x) + (1 - \alpha)f(y) \geq f(z) + \langle \nabla f(z), \alpha x + (1 - \alpha)y - z \rangle.$$

Hence, f is convex.

For the strict convexity, we repeat the same process with the strict inequality. ■

If at some point x^* we have $\langle \nabla f(x^*), z - x^* \rangle \geq 0$ for all z , then x^* is a minimum point. This is also a necessary condition: assume x^* minimum point for f over C and, by contradiction, that there exists $z \in C$ s.t. $\langle \nabla f(x^*), z - x^* \rangle < 0$, but then

$$\frac{\partial}{\partial h} f(x^* + h(z - x^*))|_{h=0} = \langle \nabla f(x^*), z - x^* \rangle < 0$$

so f decreases in the direction $z - x^*$, which contradicts the fact that x^* is a minimum.

We summarize this fact in the following proposition, which proof what we just did.

PROPOSITION 2 (Book 1.1.8). *Let $C \subseteq \mathbb{R}^n$ be convex, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable on a convex open set $A \subseteq C$. Then a point $x^* \in C$ is a minimum for f if and only if*

$$\langle \nabla f(x^*), z - x^* \rangle \geq 0 \quad \forall z \in C.$$

THEOREM 3 (Projection theorem, Book 1.1.9). *Let C be a nonempty convex closed subset of $\mathbb{R}^n$, $z \in \mathbb{R}^n$. Then there exists a unique vector $x^* \in C$ (the projection of z on C) that minimizes the function $f(x) = \|x - z\|$ over C .*

Furthermore, the point x^ is the projection of z onto C if and only if*

$$\langle z - x^*, x - x^* \rangle \leq 0 \quad \forall x \in C.$$

Proof. Optimizing the function $f(x) = \|z - x\|$ is the same as optimizing the function $f(x) = \frac{1}{2}\|z - x\|^2$. Then f is now differentiable, and $\nabla f(x) = x - z$.

$$\begin{aligned} x^* \text{ is minimum} &\iff \langle \nabla f(x^*), x - x^* \rangle \geq 0 \quad \forall x \in C \\ &\iff \langle -x^* + z, x - x^* \rangle \leq 0. \end{aligned}$$

Does such an x^* exist? We observe that

$$\min_{x \in C} f(x) = \min_{x \in C \cap \{y: \|y-z\| \leq \|z-w\|\}, w \in C} f(x).$$

We notice that we are optimizing a continuous function on a closed set, as we are optimizing on $C \cap \overline{B(z, \|z-w\|)}$, so, by Weierstrass there exists a minimum. This is also unique, indeed, let x_1^*, x_2^* be two minima, then

$$\begin{aligned} \langle z - x_1^*, x_2^* - x_1^* \rangle &\leq 0 \quad \forall z \in C \\ \langle z - x_2^*, x_2^* - x_1^* \rangle &\leq 0 \quad \forall z \in C \end{aligned}$$

Taking the sum and using the linearity of the scalar product, we obtain

$$\langle x_1^*, x_2^*, z - x_2^* - z + x_1^* \rangle \leq 0 \implies \|x_1^* - x_2^*\| \leq 0 \implies x_1^* = x_2^*.$$

■

Convex hulls

The idea is to "convexify" sets that are not convex. Let X be a set. The *convex hull* of X $\text{conv}(X)$ = intersection of all convex sets containing X . A convex combination of elements of X is a vector in the form $\sum_{i=1}^m \alpha_i x_i$, where x_i are points in X , $\alpha_i \geq 0$ and $\sum_i \alpha_i = 1$.

Remark 4. If X is convex, any convex combination of elements of X belongs to X .

Remark 5.

$$\text{conv}(X) = \left\{ z = \sum_{i=1}^m \alpha_i x_i : \alpha_i \geq 0, x_i \in X, \sum_i \alpha_i = 1, \forall m > 0 \right\}.$$

Remark 6. If $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear transformation, then $\text{conv}(Ax) = A \text{conv}(X)$.

Remark 7. $\text{conv}\left(\sum_i X_i\right) = \sum_i \text{conv}(X_i)$.

DEFINITION 8 (Affine hull). Let $X \subseteq \mathbb{R}^n$. The *affine hull* of X , denoted with $\text{aff}(X)$ is the intersection of all affine sets containing X .

Recall that M is an affine set if M is in the form $M = \bar{x} + S$, where S is a linear subspace of $\mathbb{R}^n$.

Notice that $\text{conv}(X) \subseteq \text{aff}(X)$, and intersection of affine spaces is still an affine space, so $\text{aff}(X)$ is an affine space, so $\text{aff} X = \bar{x} + S$ for some $\bar{x} \in \mathbb{R}^n$ and S subspace. We define the dimension of $\text{aff}(X)$ as the dimension of S . S is also called the parallel subspace of the affine space.

Remark 9. $\text{aff}(X) = \text{aff}(\text{conv}(X)) = \text{aff}(\text{cl}(X))$ where $\text{cl}(X)$ denotes the closure of X .

DEFINITION 10 (Cone generated by X). Let $X \subseteq \mathbb{R}^n$. We define the *cone generated by X* the set

$$\text{cone}(X) = \left\{ \sum_{i=1}^m \alpha_i x_i : \alpha_i \geq 0, x_i \in X, m = 0 \right\}.$$

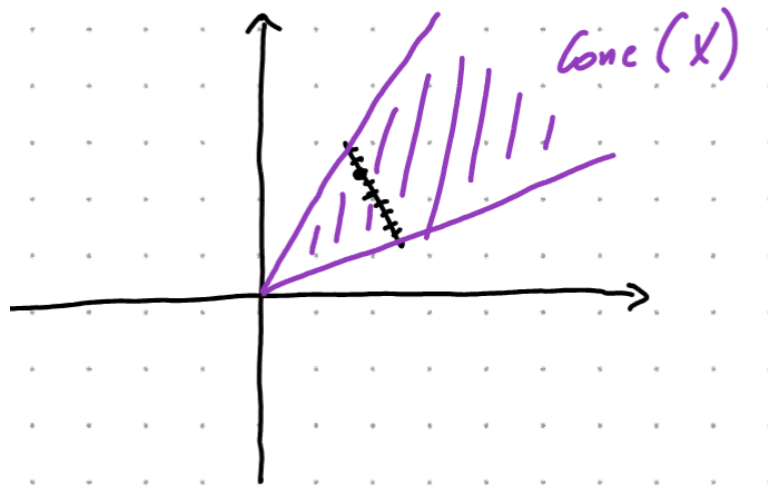


Figure 4.1: X (black) and cone generated by X (purple)

CHAPTER 5

LECTURE 5 OF 17/03/2026

To be done.

LECTURE 6 OF 18/03/2026

THEOREM 1 (Charateodory, Book 1.2.1). *Let $X \subseteq \mathbb{R}^n$ be nonempty. Then*

- a) every nonzero vector from $\text{cone}(X)$ can be represented as a positive combination of linearly independent vectors in X ;*
- b) every vector from $\text{conv}(X)$ can be represented as a convex combination of no more than $n + 1$ vectors in X .*

Proof. We prove a) and b) separately.

a): Let $y \in \text{cone}(X) \implies y = \sum_{i=1}^m \alpha_i x_i$, where $\alpha_i > 0$, $x_i \in X$. We can assume m to be the smallest index such that this representation holds true. We have to prove that $x_1, \dots, x_m$ to be linearly independent. By contraddiction, assume them dependent, so there exists $\lambda_1, \dots, \lambda_m \in \mathbb{R}$ such that $\sum_{i=1}^m \lambda_i x_i = 0$. Without loss of generality, assume at least one $\lambda_i > 0$. Define $\bar{\gamma} = \max \{ \gamma : \alpha_i - \gamma \lambda_i \geq 0 \} < +\infty$. Observe that we can write $y = \sum_i \alpha_i x_i + \underbrace{\bar{\gamma} \sum_i \lambda_i x_i}_{=0} = \sum_{i=1}^m (\alpha_i - \bar{\gamma} \lambda_i) x_i$. Then at least one of the last coefficients is zero, which is in contraddiction with the minimality of m .

b): Consider the set $Y = \{(y, 1) : y \in X\} \subseteq \mathbb{R}^{n+1}$. Let $x \in \text{conv}(X) \implies x = \sum_{i=1}^m \gamma_i x_i$, $\sum_i \gamma_i = 1$. The point $(x, 1) \in \text{cone}(Y) \implies \sum_i \gamma_i (x_i, 1) = (x, 1)$. By point a), $(x, 1) = \sum_{i=1}^m \alpha_i (\tilde{x}_i, 1)$, with $(\tilde{x}_i, 1)$ linearly independent in Y . Hence $x = \sum_{i=1}^m \alpha_i \tilde{x}_i$ with $\sum_i \alpha_i = 1$. From the fact that $(\tilde{x}_i, 1)$ are independent in Y , we conclude that $m \leq n + 1$. ■

PROPOSITION 2. Let $X \subseteq \mathbb{R}^n$ be compact (closed and bounded). Then $\text{conv}(X)$ is compact.

Proof. It is easy to show that $\text{conv}(X)$ is bounded. Since X is bounded, $\exists R > 0$ such that $\|x\| \leq R$ for any $x \in X$. Then $x \in \text{conv}(X) \iff x = \sum_i \gamma_i x_i \implies \|x\| \leq \sum_i \gamma_i \|x_i\| \leq \sum_i \gamma_i R = R$.

Closeness: Take a sequence $\{x_k^j\}_k \subseteq \text{conv}(X)$ and assume $x_k \rightarrow \bar{x}$ as $k \rightarrow +\infty$. We want to show that $\bar{x} \in \text{conv}(X)$. Each x^k can be written as

$$x_k = \sum_{i=1}^m \alpha_i^k x_i^k$$

by the Carathéodory's theorem, with $\sum_i \alpha_i = 1, \alpha_i \geq 0, x_i \in X$. Consider the sequence $\{(\alpha_1^k, \dots, \alpha_{n+1}^k, x_1^k, \dots, x_{n+1}^k)\} \subseteq \mathbb{R}^{n+1+n(n+1)} = \mathbb{R}^{(n+1)^2}$. This is a bounded sequence and converges (up to subsequences by BW) to $(\alpha_1, \dots, \alpha_{n+1}, x_1, \dots, x_{n+1})$. Observe that, since $\sum_i \alpha_i^k = 1 \implies \sum_i \alpha_i = 1, \alpha_i \geq 0$. Observe furthermore that

all $x_i \in X$ by closeness of X , so $\sum_i \alpha_i^k x_i^k \rightarrow \sum_i \alpha_i x_i$, hence $\bar{x} = \sum_{i=1}^{n+1} \alpha_i x_i$, hence $\bar{x} \in \text{conv}(X)$.

■

Relative interiors

If we consider a convex set in $\mathbb{R}^n$ which lives on an m -dimensional variety, with $m < n$ (as in 6.1), we get that the interior points of that convex sets forms the emptyset.

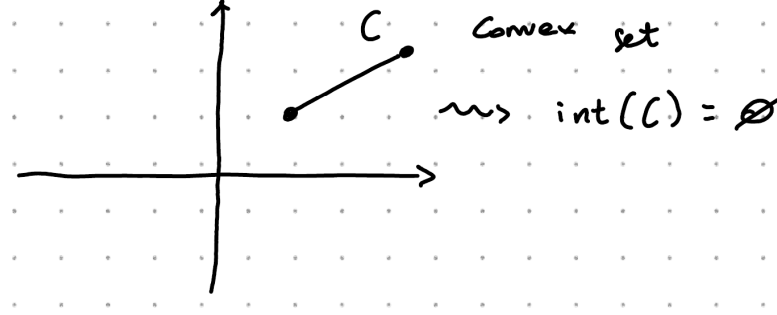


Figure 6.1: Example of emptiness of interior

DEFINITION 3 (Relative interior point). Let $C \subseteq \mathbb{R}^n$ be a convex set, $c \in C$. We say that x is a *relative interior point* if there exists an $r > 0$ such that $B(x, r) \cap \text{aff}(C) \subseteq C$.

The *relative interior* is the set of all the relative interior points, and we denote it with $\text{ri}(C)$.

Furthermore we say that C is *relatively open* if $C = \text{ri}(C)$.

PROPOSITION 4 (Line-segment principle (LSP), Book 1.3.1). Let C be a nonempty convex set. If $x \in \text{ri}(C)$ and $\bar{x} \in \text{cl}(C)$, then all points on the line segment connecting x and $\bar{x}$, except possibly $\bar{x}$ belongs to $\text{ri}(C)$.

Proof. Assume $\bar{x} \in C \implies x \in \text{ri}(C)$, so there exists $r > 0$ such that $B(x, r) \cap \text{aff}(C) \subseteq C$. Let $\alpha \in (0, 1)$ and set $x_\alpha = \alpha x + (1 - \alpha)\bar{x}$. We have to prove that $x_\alpha \in \text{ri}(C)$ for all $\alpha \in (0, 1)$.

Consider $B(x_\alpha, \alpha r)$, we have to show that $B(x_\alpha, \alpha r) \cap \text{aff}(C) \subseteq C$. Pick $y_\alpha \in B(x_\alpha, \alpha r) \cap \text{aff}(C)$. Observing that y_α belongs both to $B(x_\alpha, \alpha r)$ and $\text{aff}(C)$, in particular there exists $z_\alpha \in \mathbb{R}^n$, $\|z_\alpha\| \leq \alpha r$ such that $y_\alpha = x_\alpha + z_\alpha$, hence $y_\alpha = \alpha x + (1 - \alpha)\bar{x} + z_\alpha = \alpha \underbrace{\left(x + \frac{z_\alpha}{\alpha}\right)}_{\in B(x, r)} + \underbrace{(1 - \alpha)\bar{x}}_{\in C}$.

Observe that $x + \frac{z_\alpha}{\alpha} \in \text{aff}(C)$. Indeed $x \in \text{aff}(C)$. Consider $z_\alpha = y_\alpha - x_\alpha$, then $x_\alpha \in C \implies x_\alpha \in \text{aff}(C)$, while $y_\alpha \in \text{aff}(C)$. Putting all together, $z_\alpha \in$ parallel space to $\text{aff}(C)$, so we remain in the affine space, as we are summing a point in the affine space and a point in the parallel subspace. Moreover $x + \frac{z_\alpha}{\alpha} \in B(x, r)$ as $\|\frac{z_\alpha}{\alpha}\| \leq \frac{\alpha r}{\alpha} = r$, hence $x + \frac{z_\alpha}{\alpha} \in C$.

Consider now the case $\bar{x} \in \text{cl}(C) \setminus C$. Then there exists a sequence $\{x_k\}_k$ such that $x_k \rightarrow \bar{x}$ and $x_k \in C$ for all k . Apply then the previous argument to all x_k . Define $x_{k,\alpha} = \alpha x + (1 - \alpha)x_k \implies B(x_{k,\alpha}, r\alpha) \cap \text{aff}(C) \subseteq C$ for all k and $\alpha \in (0, 1)$. Then the sequence $x_{k,\alpha} \rightarrow x_\alpha = \alpha x + (1 - \alpha)\bar{x}$. For k large enough, $B(x_\alpha, \frac{\alpha r}{2}) \subseteq B(x_{k,\alpha}, r\alpha)$. Take the intersection with the affine hull, hence $B(x_\alpha, \frac{\alpha r}{2}) \cap \text{aff}(C) \subseteq B(x_{k,\alpha}, r\alpha) \cap \text{aff}(C)$, showing that $x_\alpha \in \text{ri}(C)$. ■

LECTURE 7 OF 24/03/2026

PROPOSITION 1 (Nonemptiness of relative interiors). *Let $C \subseteq \mathbb{R}^d$ be convex and nonempty. Then:*

- a) *the set $\text{ri}(C)$ is a nonempty convex set, $\text{aff}(\text{ri}(C)) = \text{aff}(C)$;*
- b) *if $m = \dim_{\mathbb{R}}(\text{aff}(C))$, $m > 0$ then there exists $m + 1$ vectors $x_0, x_1, \dots, x_m \in \text{ri}(C)$ such that*

$$\text{aff}(C) = \text{span}\{x_1 - x_0, x_2 - x_0, \dots, x_m - x_0\}.$$

Proof. The fact that $\text{ri}(C)$ is convex is a consequence of LSP. We have to show that $\text{ri}(C)$ is nonempty. Without loss of generality, we can assume $0 \in C$ (up to a traslation), consider $\text{aff}(C)$. If $m = 0$ then $C = \{0\} \implies \text{ri}(C) = C$ as for any $r > 0$, $B(0, r) \cap \text{aff}(C) = C$. Assume now $m > 0$. Consider $z_1, \dots, z_m \in C$ linearly independent such that $\text{aff}(C) = \text{span}\{z_1, \dots, z_m\}$. Consider the set

$$X = \left\{ x : x = \sum_{i=1}^m \alpha_i z_i, \alpha_i > 0, \sum_i \alpha_i < 1 \right\}.$$

The strategy now is to prove that $X \subseteq \text{ri}(C)$. Observe that $X \subseteq C$. Fix $\bar{x} \in X$, $x \in \text{aff}(C)$ and observe that

$$\begin{aligned} \bar{x} &= Z\bar{\alpha}, \bar{\alpha} \in \mathbb{R}^m \\ x &= Z\alpha, \alpha \in \mathbb{R}^m \end{aligned}$$

with Z the matrix with columns $z_1, \dots, z_m$. Notice that

$$\begin{aligned} \|x - \bar{x}\|^2 &= \langle Z(\alpha - \bar{\alpha}), Z(\alpha - \bar{\alpha}) \rangle \\ &= (\alpha - \bar{\alpha})^\top Z^\top Z(\alpha - \bar{\alpha}) \\ &\leq \gamma \|\alpha - \bar{\alpha}\|^2. \end{aligned}$$

Since $\bar{x} \in X$, the vector $\bar{\alpha} \in A = \{\alpha \in \mathbb{R}^m : \sum_i \alpha_i < 1, \alpha_i > 0\}$ which is an open set. If x is sufficiently close to $\bar{x}$, then α is close to $\bar{\alpha}$ by the above estimate on the norm. This implies that $\alpha \in A$, hence $x \in X$, so $B(\bar{x}, r) \cap \text{aff}(C) \subseteq X \subseteq C$. This shows that $X \subseteq \text{ri}(C)$.

Notice that, by construction, $\text{aff}(X) = \text{aff}(C)$, hence $\text{aff}(X) \subseteq \text{aff}(\text{ri}(C)) \subseteq \text{aff}(C)$.

b): fix $x_0 \in \text{ri}(C)$ (x_0 exists by a)), translate C to $C - x_0$, so that $0 \in C - x_0$. fix $z_1, \dots, z_m \in C - x_0$ linearly independent such that $\text{aff}(C - x_0) = \text{span}\{z_1, \dots, z_m\}$. Let $\alpha \in (0, 1)$ and consider $(1 - \alpha) \cdot 0 + \alpha z_i \in \text{ri}(C - x_0)$ by LSP as $0, z_i \in \text{ri}(C - x_0)$. Hence $\alpha z_i \in \text{ri}(C - x_0)$. It follows that the vectors $x_i = x_0 + \alpha z_i$ are such that $x_1 - x_0, \dots, x_m - x_0$ spans the parallel space. ■

LEMMA 2 (Prolongation lemma, Book 1.3.3). *Let $C \subseteq \mathbb{R}^d$ convex nonempty. A vector $x \in C$ belongs to $\text{ri}(C)$ if and only if for any $\bar{x} \in C$ there exists a $\gamma > 0$ such that $x + \gamma(x - \bar{x}) \in C$, i.e. every line segment connecting x to any $\bar{x}$ can be prolonged beyond x without leaving C .*

Proof. " $\implies$ " easy.

Let x satisfy $(*)$, fix $\bar{x} \in \text{ri}(C)$ (it exists thanks to the above proposition). If $x = \bar{x}$ then $x \in \text{ri}(C)$ and we are happy. If not, there exists a $\gamma > 0$ such that $z = x + \gamma(x - \bar{x}) \in C$ so that x lies within the line segment connecting $\bar{x}$ and z . By LSP, it follows that $x \in \text{ri}(C)$. ■

PROPOSITION 3 (Minima of concave functions). *Let $X \subseteq \mathbb{R}^n$ be convex and nonempty. Let $f : X \rightarrow \mathbb{R}$ be concave. Define*

$$X^* = \arg \min_{x \in X} f(x).$$

If $X^ \cap \text{ri}(X) \neq \emptyset$, then f is constant.*

Proof. Assume $x^* \in X^* \cap \text{ri}(X)$, by prolongation lemma, there exists $\gamma > 0$ such that the point $\hat{x} = x^* + \gamma(x^* - x)$ belongs to X , implying that

$$x^* = \frac{1}{\gamma + 1} \hat{x} + \frac{\gamma}{\gamma + 1} x.$$

By concavity of f ,

$$\begin{aligned} f(x^*) &= f\left(\frac{1}{\gamma+1}\hat{x} + \frac{\gamma}{\gamma+1}x\right) \\ &\geq \frac{1}{\gamma+1}f(\hat{x}) + \frac{\gamma}{\gamma+1}f(x). \end{aligned}$$

Since x^* is a minimum point, it must be that $f(y) \geq f(x^*)$ for any y , in particular, $f(\hat{x}) \geq f(x^*)$ and $f(x) \geq f(x^*)$, hence $f(x) = f(x^*)$ for any x . ■

An immediate consequence is that if $f(x) = a^\top x$ with $a \in \mathbb{R}^n$, then the minima of this function is attained at the boundary of its domain (which is assumed to be convex).

PROPOSITION 4 (Continuity of convex functions, Book 1.3.11). *If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, then f is continuous. More generally, if $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ is proper and convex, then the restriction of f to $\text{dom}(f)$ is continuous in $\text{ri}(\text{dom}(f))$.*

Proof. Assume $0 \in \text{dom}(f)$ (if not translate). Without loss of generality, assume that the set $X = \{x : \|x\|_\infty \leq 1\} \subseteq \text{dom}(f)$ (up to dilatation). It is now sufficient to prove that f is continuous at 0: for any sequence $\{x_k\}_k$ such that $x_k \rightarrow 0$ we want to prove $f(x_k) \rightarrow f(0)$. Define e_i , for $i = 1, \dots, 2^n$ the corners of X . Let $x \in X$, then

$$x = \sum_{i=1}^{2^n} \alpha_i e_i \quad \sum_i \alpha_i = 1, \alpha_i \geq 0.$$

Then

$$\begin{aligned} f(x) &= f\left(\sum_{i=1}^{2^n} \alpha_i e_i\right) \\ &\leq \sum_i \alpha_i f(e_i) \\ &\leq \sum_i \alpha_i \left(\max_j f(e_j)\right) \\ &\leq \max_j f(e_j). \end{aligned}$$

Consider now the sequence $x_k \rightarrow 0$, define $y_k = x_k / \|x_k\|$, $z_k = -x_k / \|x_k\|$. We can see x_k as a convex combination of y_k and 0, hence

$$\begin{aligned} f(x_k) &= f(\|x_k\|_\infty y_k - (1 - \|x_k\|_\infty) \cdot 0) \\ &\leq \|x_k\| f(y_k) + (1 - \|x_k\|) f(0). \end{aligned}$$

Observe that $\|x_k\| \rightarrow 0$, and $f(y_k)$ is bounded ($y_k \in \text{dom}(f)$), hence $\|x_k\| f(y_k) \rightarrow 0$ as $k \rightarrow +\infty$. This shows that

$$\limsup_k f(x_k) \leq f(0).$$

For z_k instead:

$$f(0) = f\left(\frac{1}{1 + \|x_k\|} x_k + \frac{\|x_k\|}{1 + \|x_k\|} z_k\right) \leq \liminf_k f(x_k).$$

Hence $f(0) \leq \liminf_k f(x_k) \leq \limsup_k f(x_k) \leq f(0)$. This shows that f is continuous at 0. ■

7.1 INTERMEZZO

Let $E \subseteq \mathbb{R}^{n+1}$, $f : D \rightarrow \overline{\mathbb{R}}$ where $D = \{x \in \mathbb{R}^n : \exists w \in \mathbb{R} \text{ s.t. } (x, w) \in E\}$, $f(x) = \inf_w (x, w) \in E$.

Let now $f : X \rightarrow \overline{\mathbb{R}}$ any function. We define the *closure of f* as

$$\text{cl}(f)(x) = \inf \{w : (x, w) \in \text{cl}(\text{epi}(f))\}.$$

The *convex closure* of f is

$$\check{\text{cl}}(f)(x) = \inf \{w : (x, w) \in \text{cl}(\text{conv}(\text{epi}(f)))\}.$$

It can be proven that $\check{\text{cl}}(f) = \text{cl}(F)$ where

$$F(x) = \inf \{w : (x, w) \in \text{conv}(\text{epi}(f))\}.$$

PROPOSITION 1 (Book 1.3.13). *Let $f : X \rightarrow \overline{\mathbb{R}}$. Then*

$$\inf_{x \in X} f(x) = \inf_{x \in \mathbb{R}^n} \text{cl}(f)(x) = \inf_{x \in \mathbb{R}^n} F(x) = \inf_{x \in \mathbb{R}^n} \check{\text{cl}}(f)(x).$$

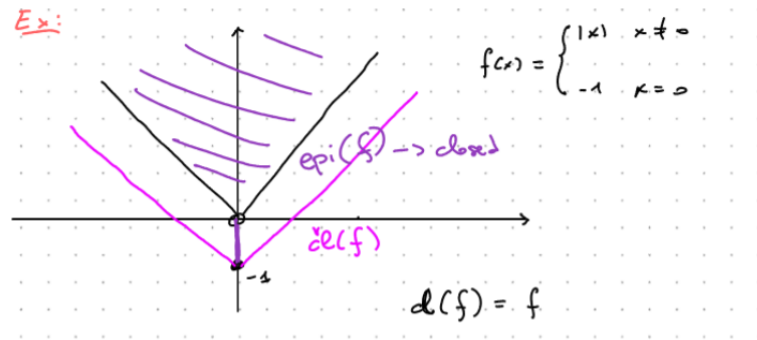


Figure 7.1: Example of convex closure of a function

Proof. If $\text{epi}(f) = \emptyset$, then $f \cong +\infty$, so this case is verified. Suppose now $\text{epi}(f) \neq \emptyset$. Let $f^* = \inf_{x \in \mathbb{R}^n} \text{cl}(f)(x)$. for any sequence $\{(\bar{x}_k, \bar{w}_k)\}_k \subseteq \text{cl}(\text{epi}(f))$ with $\bar{w}_k \rightarrow f^*$ we can construct a subsequence $\{(x_k, w_k)\}_k \subseteq \text{epi}(f)$ such that $w_k \rightarrow f^*$, $|w_k - \bar{w}_k| \rightarrow 0$. For each $x_k \in X$, it holds $(x_k, w_k) \in \text{epi}(f)$, then $f(x_k) \leq w_k$. But then

$$\inf_{x \in X} f(x) \leq \limsup_k f(x_k) \leq \lim_k w_k = f^* \leq \text{cl}(f)(x) \leq f(x).$$

Hence

$$\inf_{x \in X} f(x) = \inf_{x \in \mathbb{R}^n} \text{cl}(f)(x).$$

The next part uses the same argument. ■

LECTURE 8 OF 25/03/2026

PROPOSITION 1. *Let $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ a function. Then*

- a) $\text{cl}(f)$ is the greatest closed function below f (recall that we say $g \leq f$ if $g(x) \leq f(x)$ for any x);*
- b) $\check{\text{cl}}(f)$ is the greatest convex closed function below f .*

Proof. Let $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ closed below f , then $\text{epi}(f) \subseteq \text{epi}(g)$, but $\text{epi}(\text{cl}(f)) = \text{cl}(\text{epi}(f)) = \text{intersection of all closed sets containing } \text{epi}(f)$. Notice that $\text{epi}(g)$ is closed, as g is closed. Hence $\text{cl}(\text{epi}(f)) \subseteq \text{epi}(g)$ as $\text{epi}(g)$ is one of the closed sets containing $\text{epi}(f)$. This shows that $\text{cl}(f) \leq g$. The same argument works for b). ■

8.1 RECESSON CONES

DEFINITION 1 (Recession direction/cone). Let C be convex and nonempty. We say that a direction $d \in \mathbb{R}^n$ is a *recession direction* for C if $x + \alpha d \in C$ for any $x \in C, \alpha \geq 0$.

We denote with R_C the *recession cone* of C the set of all recession directions.

Example 2.

$$C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}. \quad R_C = \{0\}. \quad (8.1)$$

$$C = (-\infty, 2]. \quad R_C = (-\infty, 0]. \quad (8.2)$$

PROPOSITION 3 (Recession cone theorem 1.4.1). *Let C closed convex nonempty. Then*

- a) R_C is closed and convex;*
- b) $d \in R_C \iff$ there exists $x \in C$ such that $x + \alpha d \in C$ for all $\alpha \geq 0$.*

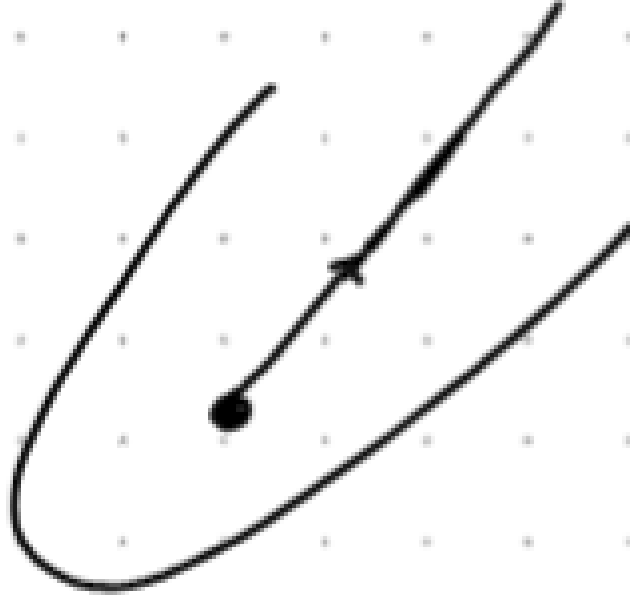


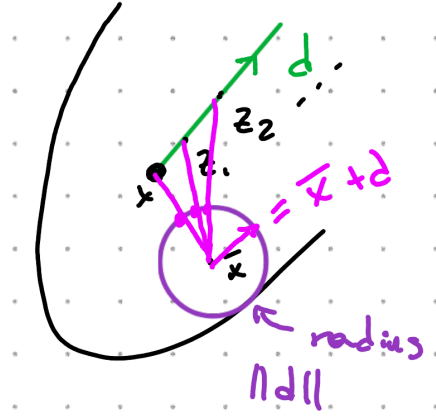
Figure 8.1: Example of recession cone

Proof. We first prove convexity. Let $d_1, d_2 \in R_C$, $\gamma_1, \gamma_2 \in [0, 1]$ such that $\gamma_1 + \gamma_2 = 1$. Fix any $x \in C$, $\alpha \geq 0$ and consider $x + \alpha(\gamma_1 d_1 + \gamma_2 d_2) = x(\gamma_1 + \gamma_2) + \alpha(\gamma_1 d_1 + \gamma_2 d_2) = \gamma_1(x + \alpha d_1) + \gamma_2(x + \alpha d_2)$ which is a convex combination of points in C . For the closeness, consider $\{d_k\}_k$ a sequence in R_C such that $d_k \rightarrow d \in \mathbb{R}^n$. Then for any $x \in C$ and for any $\alpha \geq 0$, $x + \alpha d_k \in C \rightarrow x + \alpha d$ as $k \rightarrow +\infty$. This shows that $x + \alpha d \in C$ as C is closed.

For point b), the implication " $\implies$ " is obvious.

Let $d \in \mathbb{R}^n$ and assume $x \in C$ such that $x + \alpha d \in C$ for any $\alpha \geq 0$. Let $\bar{x} \in C$. We have to prove $\bar{x} + \alpha d \in C$. Assume, without loss of generality, $\alpha = 1$ as we can retrieve the general case with a scaling argument. Define a sequence $z_k = x + kd$, $z_k \in C$ for all k . If $\bar{x} = z_k$ for some k then we are done. Assume then $\bar{x} \neq z_k$ and define $d_k = \frac{z_k - \bar{x}}{\|z_k - \bar{x}\|} \|d\|$. Then

$$\begin{aligned} d_k &= \frac{z_k - \bar{x} + x - x}{\|z_k - \bar{x}\|} \|d\| = \frac{\overbrace{z_k - x}^{=kd}}{\|z_k - \bar{x}\|} \|d\| + \frac{x - \bar{x}}{\|z_k - \bar{x}\|} \\ &= \frac{k\|d\|}{\|z_k - \bar{x}\|} d + \frac{x - \bar{x}}{\|z_k - \bar{x}\|}. \end{aligned}$$

Figure 8.2: Visual intuition on the sequence z_k

The last term goes to 0, indeed, $x - \bar{x}$ is fixed, $\|d\|$ is fixed and $\|z_k\| \nearrow +\infty$ as $k \nearrow +\infty$. I claim that the first term goes to d : in fact,

$$\frac{k\|d\|}{\|x + kd - \bar{x}\|} = \frac{k\|d\|}{k\|d + (x - \bar{x})/k\|} \rightarrow 1.$$

This shows that $d_k \rightarrow d$ as $k \nearrow +\infty$. For k large enough, the vector $\bar{x} + dk$ belongs to the segment connecting $\bar{x}$ to z_k , so, by convexity of C , $\bar{x} + d_k \in C$, hence $\bar{x} + d \in C$ as C is closed. ■

PROPOSITION 4 (Properties of the recession cone). *Let $C \subseteq \mathbb{R}^n$ closed convex nonempty. Then*

- a) R_C contains a nonzero direction if and only if C is unbounded;
- b) $R_C = R_{\text{ri}(C)}$
- c) if $\{C_i\}_{i \in I}$ is a collection of closed convex sets such that $\bigcap_{i \in I} C_i \neq \emptyset$, then $R_{\bigcap_{i \in I} C_i} = \bigcap_{i \in I} R_{C_i}$;
- d) fix $W \subseteq \mathbb{R}^n$ compact, $A \in \mathbb{R}^{n \times n}$ a matrix. Let $V = \{x \in C : Ax \in W\}$. Assume V nonempty. Then $R_V = R_C \cap N(A)$ where $N(A) = \{x \in \mathbb{R}^n : Ax = 0\}$ is the null space of the linear operator A .

Proof. Not required. ■

Example 5. The following example shows how the closeness of C can play a role in the recession cone. Consider the set

$$C = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, y \geq 0\} \cup \{(1, 0)\}.$$

Then $R_C = \{0\}$, indeed, if we start from the point $(1, 0)$, we cannot move along any direction without leaving C , while $R_{\text{ri}(C)} = \text{span}\{(0, 1)\}$.

8.2 LINEARITY SPACE

We define the linearity space for a set C as the set

$$L_C = R_C \cap (-R_C).$$

In other words, $d \in L_C$ if and only if $x + \alpha d \in C$ and $x - \alpha d$ are inside C for any $\alpha \geq 0$.

PROPOSITION 1 (Properties of L_C). *Let $C \subseteq \mathbb{R}^n$ convex closed nonempty, then*

1. L_C is a subspace;
2. $L_C = L_{\text{ri}(C)}$;
3. same as above.

Proof. Not required. ■

Example 2 (Important!). Consider the set

$$C = \{x \in \mathbb{R}^n : x^\top Qx + c^\top x + b \leq 0\}$$

where $Q \in \mathbb{R}^{n \times n}$ is SPD, $c \in \mathbb{R}^n$, $b \in \mathbb{R}$. We want to compute R_C . Let $x \in C$, $d \in \mathbb{R}^n$. Then $x + \alpha d \in C$ for any $\alpha \geq 0$ if and only if

$$\begin{aligned} (x + \alpha d)^\top Q(x + \alpha d) + c^\top (x + \alpha d) + b &\leq 0 \\ \iff \underbrace{x^\top Qx + c^\top x + b}_{\leq 0} + \alpha(c^\top d + 2x^\top Qd) + \alpha^2 d^\top Qd &\leq 0 \end{aligned}$$

for any $\alpha \geq 0$ and for any $x \in C$.

The quantity $\alpha^2 d^\top Qd \geq 0$ since Q is SPD, hence we need $d^\top Qd = 0$. Since Q is SPD, we can compute its Cholesky factorization, hence $Q = M^\top M \implies d^\top M^\top M d = 0$, so $Md = 0$.

8.2 Linearity space

Furthermore, we need also $\alpha c^\top d \leq 0$. Finally, the recession cone turns out to be

$$R_C = \left\{ d \in \mathbb{R}^n : Qd = 0, c^\top d \leq 0 \right\}.$$

Moreover,

$$L_C = \left\{ d \in \mathbb{R}^n : Qd = 0, c^\top d = 0 \right\}.$$

LECTURE 9 OF 31/03/2026

PROPOSITION 1. Let $C \subseteq \mathbb{R}^n$ nonempty convex, let $S \subseteq L_C$ be a subspace. Then we can decompose C as

$$C = S + (C \cap S^\perp).$$

Proof. Not required. ■

PROPOSITION 2 (Book 1.4.5). Let $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ be closed proper convex function. Let $V_\gamma = \{x \in \mathbb{R}^n : f(x) \leq \gamma\}$ for all $\gamma \in \mathbb{R}$ (sublevel sets of f). Then

a) all nonempty V_γ have the same recession cone, denoted by

$$R_f = \left\{ d \in \mathbb{R}^n : (d, 0) \in R_{\text{epi}(f)} \right\}$$

b) if one V_γ is compact, then all the sublevel sets are compact.

Proof. Fix $\gamma \in \mathbb{R}$, assume $V_\gamma \neq \emptyset$. Define the γ -slice $S = \{(x, \gamma) : f(x) \leq \gamma\}$. Observe that $S = \text{epi}(f) \cap \{(x, \gamma) : x \in \mathbb{R}^n\}$. We already know $\text{epi}(f)$ closed convex, so

$$R_S = R_{\text{epi}(f)} \cap R_{\{(x, \gamma) : x \in \mathbb{R}^n\}} = R_{\text{epi}(f)} \cap \{(d, 0) : d \in \mathbb{R}^n\} = \left\{ (d, 0) : d \in R_{\text{epi}(f)} \right\}$$

which does not depend on γ .

Point b): from a), we know $R_{V_{\gamma_1}} = R_{V_{\gamma_2}}$ for any $\gamma_1, \gamma_2 \in \mathbb{R}$ such that their sublevel sets are nonempty. If, for $\bar{\gamma} \in \mathbb{R}$, $V_{\bar{\gamma}}$ is compact, then $R_{V_{\bar{\gamma}}} = \{0\}$. Then for a) every recession cone is equal, hence $R_{V_\gamma} = \{0\}$ for any γ . This shows that V_γ is compact for every $\gamma \in \mathbb{R}$. ■

Remark 3. Let $d \in R_f$ if $d \in R_{V_\gamma}$ for any γ . Fix $\gamma \in \mathbb{R}$, then $d \in R_{V_\gamma}$ implies that for $x \in V_\gamma$, $x + \alpha d \in V_\gamma$ for any $\alpha \geq 0$, hence $f(x + \alpha d) \leq \gamma$ for every $\alpha \geq 0$. This shows that on the direction d , the function f cannot increase, so d is a direction of continuous non-ascent for f .

DEFINITION 4 (Linearity space of a function). The linearity space of a function is defined in the same way as the linearity space of a set.

Remark 5. If $d \in L_f$, then

$$\begin{aligned} f(x + \alpha d) &\leq f(x) \quad \forall \alpha \geq 0 \\ f(x - \alpha d) &\leq f(x) \quad \forall \alpha \geq 0. \end{aligned}$$

I claim that d is a constancy direction for f . Let $x = \frac{1}{2}(x - \alpha d) + \frac{1}{2}(x + \alpha d)$. Indeed, assume by contraddiction $f(x - \alpha d) < f(x)$. Then

$$\begin{aligned} f(x) &= f\left(\frac{1}{2}(x - \alpha d) + \frac{1}{2}(x + \alpha d)\right) \\ &\leq \frac{1}{2} \underbrace{f(x - \alpha d)}_{< f(x)} + \frac{1}{2} \underbrace{f(x + \alpha d)}_{\leq f(x)} < f(x). \end{aligned}$$

which is a contraddiction.

This shows that $f(x + \alpha d) = f(x - \alpha d) = f(x)$ for any $\alpha \geq 0$. The space L_f is also called *constancy space*.

Example 6. Consider $f(x) = x^\top Qx + c^\top x + b$ as before. It is not difficult to show that

$$L_f = \left\{ d : Qd = 0, c^\top d = 0 \right\}.$$

9.1 RECESSION FUNCTION

Consider $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ closed proper convex. This implies $\text{epi}(f)$ is closed convex subset of $\mathbb{R}^{n+1}$. Consider $R_{\text{epi}(f)} \subseteq \mathbb{R}^{n+1}$. Question: is that set the epigraph of something? The answer is yes, for every x , the quantity $\inf_w \left\{ (d, w) : d \in R_{\text{epi}(f)} \right\}$ is finite or $+\infty$. Since $(d, w) \in R_{\text{epi}(f)}$ then $(d, f(x)) + \alpha(d, w) \in \text{epi}(f)$, hence $(x + \alpha d, f(x) + \alpha w) \in \text{epi}(f) \implies f(x + \alpha d) \leq f(x) + \alpha w$. Assume w can be taken as negative as we want, so $f(x + d) = -\infty$, but this is not possible. This implies that $R_{\text{epi}(f)}$ can be the epigraph of some function.

DEFINITION 1 (Recession function). We define r_f the recession function of f as the function such that $\text{epi}(r_f) = R_{\text{epi}(f)}$.

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PROPOSITION 1 (Book 1.4.6). *Let $f : \mathbb{R} \rightarrow (-\infty, +\infty]$ closed proper convex, Consider R_f and L_f the recession cone and constancy space. Then*

$$\begin{aligned} R_f &= \{d \in \mathbb{R}^n : r_f(d) \leq 0\} \\ L_f &= \{d \in \mathbb{R}^n : r_f(d) = 0\} \end{aligned}$$

Proof. It follows from the definition of recession cone of a function that

$$\begin{aligned} R_f &= \left\{ d \in \mathbb{R}^n : (d, 0) \in R_{\text{epi}(f)} \right\} \\ &= \left\{ d \in \mathbb{R}^n : (d, 0) \in \text{epi}(r_f) \right\} \\ &= \{d \in \mathbb{R}^n : r_f \leq 0\} \end{aligned}$$

For the linearity space instead, we observe that $L_f = R_f \cap (-R_f)$, $r_f(d) \leq 0$ and $r_f(-d) \leq 0$ for any $d \in L_f$. Then

$$r_f(0) = r_f\left(\frac{1}{2}d + \frac{1}{2}(-d)\right) \leq \frac{1}{2}r_f(d) + \frac{1}{2}r_f(-d).$$

Observe that $r_f(0) = 0$. Indeed, $r_f(0) = \inf \left\{ w : (0, w) \in R_{\text{epi}(f)} \right\} = 0$ because we can take w as negative as we want, but f is proper, hence that infimum is 0. It follows that $r_f(d) = r_f(-d) = 0$ for any $d \in L_f$. ■

PROPOSITION 2 (Book 1.4.7). *Let $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ be closed proper convex. Then, for every $x \in \text{dom}(f)$, for any $d \in \mathbb{R}^n$ we have*

$$r_f(d) = \sup_{\alpha > 0} \frac{f(x + \alpha d) - f(x)}{\alpha} = \lim_{\alpha \rightarrow +\infty} \frac{f(x + \alpha d) - f(x)}{\alpha}.$$

Intuitively it gives a measure on "how fast f is diverging in the direction d ".

Proof. Not required. ■

10.1 NONEMPTYNESS OF INTERSECTION OF CLOSED SETS

Now we ask ourselves a question: for $\{C_k\}_k$ a sequence of sets in $\mathbb{R}^n$ nested, i.e. $C_{k+1} \subseteq C_k$ for any k . We know that C_k is compact for any k , then $\bigcap_k C_k \neq \emptyset$. Can we do similarly also if C_k are only closed?

Why we ask such a question?

Example 1. Assume that we are looking for $\inf_{x \in X} f(x)$. Consider a sequence $\{\gamma_k\}_k \subseteq \mathbb{R}^n$, $\gamma_k \rightarrow \gamma^* = \inf_{x \in X} f(x)$. Does such a point exist? Define $X_k = \{x : f(x) \leq \gamma_k\}$. One can show that if $\bigcap_k X_k$ is nonempty then there exists x^* such that $f(x^*) = \gamma^*$.

Example 2. Consider $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ a linear map, C a closed set. Is the set AC closed? Not always. Indeed, fix a sequence $\{y_k\}_k \subseteq AC$, $y_k \rightarrow \bar{y} \in \mathbb{R}^n$. Does $\bar{y} \in C$? Set $C_k = C \cap N_k$ where $N_k = \{x : \|Ax - \bar{y}\| \leq \|y_k - \bar{y}\|\}$. One can show that if the intersection of C_k is nonempty, then there exists a point $\bar{x} \in \bigcap_k C_k$ such that $A\bar{x} = \bar{y}$.

From now on, C_k denotes a sequence of nested nonempty convex closed sets.

DEFINITION 3. We say that a sequence $\{x_k\}_k$, $x_k \neq 0$ is an *asymptotic sequence* of $\{C_k\}_k$ if $\|x_k\| \rightarrow +\infty$ and

$$\frac{x_k}{\|x_k\|} \rightarrow \frac{d}{\|d\|}$$

where $d \neq 0$, $d \in \bigcap_k R_{C_k}$. Intuitively, we can say that $\{x_k\}_k$ "explodes only in a particular direction".

Example 4. Consider $C_k = \{x : x \geq k\}$, then $d = 1 \in \bigcap_k R_{C_k}$. Moreover, any unbounded sequence in $\mathbb{R}$ is asymptotic, as $\mathbb{R}$ has a unique direction in which a sequence can escape.

Remark 5. Given any unbounded sequence $\{x_n\}_n \subseteq \mathbb{R}^d$, there exists a subsequence $\{x_{n_k}\}_k$ that is asymptotic, because every cluster point of $\{x_n / \|x_n\|\}_n$ is a common recession direction for each C_k .

DEFINITION 6 (Retractive asymptotic sequence). An asymptotic sequence $\{x_k\}_k$ of $\{C_k\}_k$, with d as asymptotic direction is *retractive* if there exists a $\bar{k}$ big enough such that $x_k - d \in C_k$ for any $k > \bar{k}$.

DEFINITION 7. A sequence $\{C_k\}_k$ is *retractive* if all its asymptotic sequence are retractive. A set $C \subseteq \mathbb{R}^n$ is retractive in the same way.

Example 8. The sequence $C_k = \{x : x \geq k\}$ is not retractive because there exist a sequence $\{x_k\}_k$ such that $x_k - d \notin C_k$, e.g. $x_k = k$, $d = 1$, so $x_k - d = k - 1 \notin C_k$.

Let $C_k = \{(x, y) \in \mathbb{R}^2 : |x| \leq \frac{1}{k}\}$. Then

$$\bigcap_k C_k = \{(0, y) : y \in \mathbb{R}\} \neq \emptyset.$$

This sequence is retractive, as the only direction of escape is y .

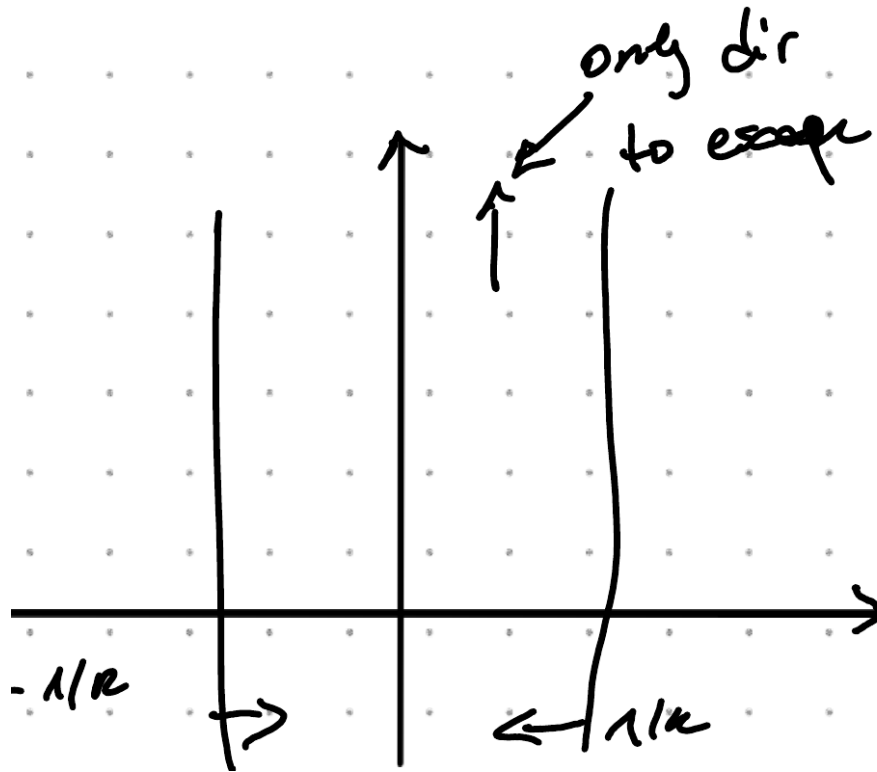


Figure 10.1: Visual intuition for the example

Remark 9. Assume $\{C_k^1\}_k, \{C_k^2\}_k, \dots, \{C_k^m\}_k$ to be m families of nested closed convex nonempty sets. Then define

$$N_k = C_k^1 \cap \dots \cap C_k^m$$

$$T_k = C_k^1 \times \dots \times C_k^m.$$

If each sequence is retractive and $N_k \neq \emptyset$ then $\{N_k\}_k$ and $\{T_k\}_k$ are retractive.

PROPOSITION 10. Polyhedral sets are retractive.

Proof. Corollary of the above remark. ■

10.1 Nonemptiness of intersection of closed sets

PROPOSITION 11. *A retractive nested sequence of nonempty closed convex sets has nonempty interior.*

Proof. Sketched in class, not reported here. ■

Example 12. Retractiveness is *not* a necessary condition: consider the sequence

$$C_k = \left\{ (x, y) \in \mathbb{R}^2 : y \geq x^2 - \frac{1}{k} \right\}.$$

Then the intersection is nonempty but C_k is not retractive: the only direction of escape is $(0, 1)$, but if we take a point on the boundary of each C_k we go out of the set.

The question now is: how can we easily prove retractiveness? The answer is in the following:

PROPOSITION 13 (Book 1.4.11). *Denote $R = \bigcap_k R_{C_k}$ and $L = \bigcap L_{C_k}$. Then*

a) If $R = L$ then $\{C_k\}_k$ is retractive. Furthermore it holds true that

$$\bigcap_k C_k = L + \tilde{C}$$

where $\tilde{C}$ is a nonempty compact set;

b) Fix $X \subseteq \mathbb{R}^n$ retractive, closed and convex. Assume that the sets $\bar{C}_k = X \cap C_k$ are nonempty and $R_X \cap R \subset L$. Then $\bar{C}_k$ is retractive.

Proof. Not required. ■

PROPOSITION 14 (Existence of solution of convex quadratic problems, book 1.4.12). *Let $Q \in \mathbb{R}^{n \times n}$ be SPD, $c \in \mathbb{R}^n$, $a_1, \dots, a_m \in \mathbb{R}^n$, $b_1, \dots, b_m \in \mathbb{R}$. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $x \mapsto x^\top Qx + c^\top x$, and $X = \left\{ x \in \mathbb{R}^n : a_j^\top x \leq b_j, \forall 1 \leq j \leq m \right\}$. Assume $\inf_{x \in X} f(x)$ to be finite. Then there exists $x^* \in X$ such that*

$$f(x^*) = \inf_{x \in X} f(x).$$

Proof. Let $f^* = \inf_{x \in X} f(x)$. Let $\{\gamma_k\}_k \subseteq \mathbb{R}$ such that $\gamma_k \rightarrow f^*$ as $k \rightarrow +\infty$. Define $\bar{C}_k = X \cap \{x : f(x) \leq \gamma_k\}$. Let R_X be the recession cone of X , $R = R_{\bigcap_k C_k}$, $L = L_{\bigcap_k C_k} (= R_{C_k} = L_{C_k})$ since we already know that the sublevel sets share all the same recession cone. In a previous example we computed

$$\begin{aligned} R &= \left\{ d : Qd = 0, c^\top d \leq 0 \right\} \\ L &= \left\{ d : Qd = 0, c^\top d = 0 \right\} \\ R_X &= \left\{ d : a_j^\top d \leq 0, \forall 1 \leq j \leq m \right\}. \end{aligned}$$

Take $d \in R_X \cap R$, assume $d \notin L$. Then $Qd = 0, c^\top d < 0, a_j^\top d \leq 0$ for all j , but then $f(x + \alpha d) \rightarrow -\infty$ as $\alpha \rightarrow +\infty$, which contradicts the fact that f^* is finite. Hence $R_X \cap R \subseteq L$, so by the previous theorem, $\{\bar{C}_k\}_k$ is retractive, and so there exists $x^* \in \bigcap_k \bar{C}_k$ such that $f(x^*) = f^*$. ■

LECTURE 11 OF 07/04/2026

PROPOSITION 1 (Book 1.4.13). *Let $X, C \subseteq \mathbb{R}^n$ convex closed nonempty. Let $A \in \mathbb{R}^{m \times n}$, $m \in \mathbb{N}$, $m > 0$ and denote with $N(A)$ the nullspace of A . If X is retractive and $R_X \cap R_C \cap N(A) \subseteq L_C$ then $A(X \cap C)$ is closed.*

Proof. Let $\{y_k\}_k \subseteq A(X \cap C)$ such that $y_k \rightarrow \bar{y}$ as $k \rightarrow +\infty$. To prove that $\bar{y} \in A(X \cap C)$, hence that there exists $\bar{x} \in X \cap C$ such that $A\bar{x} = \bar{y}$. Define $C_k = C \cap N_k$ where $N_k = \{x : \|Ax - \bar{y}\| \leq \|y_k - \bar{y}\|\}$, these C_k are convex and closed. Moreover $R_{C_k} = R_C \cap R_{N_k} = R_C \cap N(A)$. Use then point b) of proposition of the last lecture, and we are done. Why? $\|y_k - \bar{y}\| \rightarrow 0$. For every $x \in C_k$, $\|Ax - \bar{y}\| \leq \|y_k - \bar{y}\| \rightarrow 0$. For $x \in \bigcap_k C_k$, $\|Ax - \bar{y}\| \leq$ any value of a sequence that is going to 0, hence $\|Ax - \bar{y}\| = 0 \implies Ax = \bar{y}$. ■

Remark 2. If $C = \mathbb{R}^n$, then $L_C = \mathbb{R}^n$, X a polyhedral set (hence retractive), then $A(C \cap X) = AX$ is closed.

Remark 3. $X = \mathbb{R}^n$, $R_X = \mathbb{R}^n$. $A(X \cap C) = AC$ is closed if $R_C \cap N(A) = \{0\}$.

PROPOSITION 4. *Let $C_1, \dots, C_m$ be nonempty closed convex sets such that, if $d_1 + \dots + d_m = 0$ for $d_i \in R_{C_i}$ then $d_i \in L_{C_i}$, then $C_1 + \dots + C_m$ is closed.*

Proof. Not required. ■

Remark 5. If C_1 and $-C_2$ are closed and convex, then $C_1 - C_2$ is closed convex if $R_{C_1} \cap R_{C_2} = \{0\}$, so if one between C_1 and C_2 is bounded.

Example 6. Let

$$\begin{aligned} C_1 &= \{(x, y) : x \geq 0, y \geq 1/x\} \\ C_2 &= \{(x, y) : x = 0\}. \end{aligned}$$

Then

$$\begin{aligned} C_1 + C_2 &= \{(x_1, y_1) + (x_2, y_2) : (x_1, y_1) \in C_1, (x_2, y_2) \in C_2\} \\ &= \{(x_1, y_1 + \alpha) : (x_1, y_1) \in C_1, \alpha \in \mathbb{R}\} \\ &= \{(x, y) : x > 0\} \end{aligned}$$

which is an open set. Observe that

$$\begin{aligned} d_1 &= (0, 1) \in R_{C_1} \\ d_2 &= (0, -1) \in R_{C_2}. \end{aligned}$$

Then $d_1 + d_2 = 0$ and $d_2 \in L_{C_2}$ but $d_1 \notin L_{C_1}$.

11.1 HYPERPLANES

An hyperplane is a set

$$H = \{x : a^\top x = b\}$$

for some $a \in \mathbb{R}^n, a \neq 0, b \in \mathbb{R}$. For $\bar{x} \in H$, we have

$$H = \{x : a^\top x = a^\top \bar{x}\}$$

hence $H = \bar{x} + \{x : a^\top x = 0\}$, where the latter is the parallel subspace to H . Furthermore we have the closed halfspaces where we replace the equal sign with a large inequality, closed with strict.

11.1.1 Separation

We say that H separates C_1 and C_2 if $a^\top x_1 \leq b \leq a^\top x_2$ or the converse. We also write

$$\sup_{x_1 \in C_1} a^\top x_1 \leq \inf_{x_2 \in C_2} a^\top x_2.$$

Assume $\bar{x} \in \text{cl}(C)$ for $C \subseteq \mathbb{R}^n$. If H separates $\{\bar{x}\}$ from C , we say H is a supporting hyperplane at $\bar{x}$.

PROPOSITION 1 (Supporting hyperplane theorem, book 1.5.1). *Let $C \subseteq \mathbb{R}^n$ nonempty convex, $\bar{x} \in \mathbb{R}^n, \bar{x} \notin \text{int}(C)$. Then there exists an hyperplane H that passes through $\bar{x}$ and contains C in one of its closed halfspaces, i.e. there exists $a \in \mathbb{R}^n, a \neq 0$ such that*

$$a^\top \bar{x} \leq a^\top x \quad \forall x \in C.$$

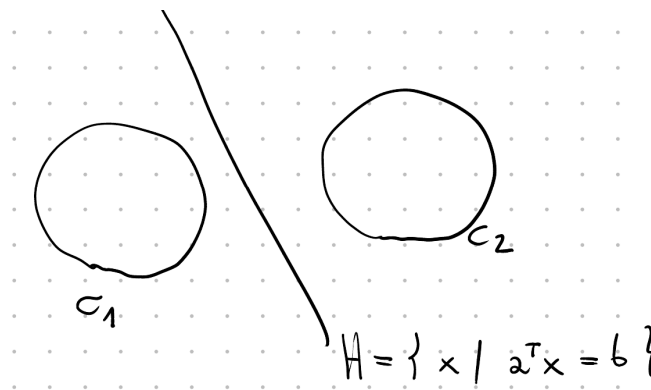


Figure 11.1: Hyperplane separating two closed convex sets

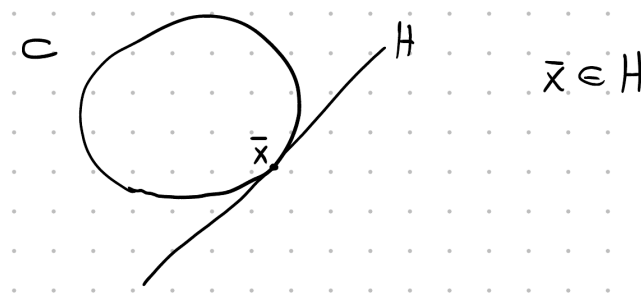


Figure 11.2: Example of supporting hyperplane

Proof. Consider $\bar{x} \notin \text{int}(C)$. One can prove that there exists a sequence $\{x_k\}_k \subseteq \mathbb{R}^n \setminus \text{cl}(C)$ such that $x_k \rightarrow \bar{x}$. (to prove this, later...)

Consider $\hat{x}_k$ the projection of x_k onto $\text{cl}(C)$ which is a closed convex set, so the projection exists and it is unique. By projection theorem, it holds that

$$\langle \hat{x}_k - x_k, x - \hat{x}_k \rangle \geq 0 \quad \forall x \in \text{cl}(C).$$

Rewrite

$$\begin{aligned} (\hat{x}_k - x_k)^\top \cdot x &\geq (\hat{x}_k - x_k)^\top \cdot (\hat{x}_k \pm x_k) \\ &= \underbrace{(\hat{x}_k - x_k)^\top \cdot (\hat{x}_k - x_k)}_{\geq 0} + (\hat{x}_k - x_k)^\top \cdot x_k \\ &\geq (\hat{x}_k - x_k)^\top \cdot x_k \end{aligned}$$

Define $a_k = \frac{\hat{x}_k - x_k}{\|\hat{x}_k - x_k\|}$, we obtain that $a_k^\top \cdot x \geq a_k^\top \cdot x_k$ for any $x \in \text{cl}(C)$

and for any k . The sequence $\{a_k\}_k \subseteq \overline{B(0,1)}$, so by BW, up to subsequences, $a_k \rightarrow \bar{a} \in \mathbb{R}^n$, $\|\bar{a}\| = 1$. Taking the limit

$$\begin{array}{ccc} a_k^\top \cdot x & \geq & \top_k x_k \\ \downarrow & & \downarrow \\ \bar{a}^\top \cdot x & \geq & \bar{a}^\top \cdot \bar{x}. \end{array}$$

Hence $\bar{a}$ defines our target hyperplane. ■

PROPOSITION 2 (Separating hyperplane theorem (finite HB), Book 1.5.2). *Let $C_1, C_2 \subseteq \mathbb{R}^n$ be nonempty convex. If C_1 and C_2 are disjoint, then there exists $a \in \mathbb{R}^n$, $a \neq 0$ such that*

$$a^\top x_1 \leq a^\top x_2 \quad \forall x_1 \in C_1 \forall x_2 \in C_2.$$

Proof. Let $C = C_1 - C_2$ the algebraic set difference. Observe that $0 \notin C$, as $C_1 \cap C_2 = \emptyset$. Let H be the hyperplane separating $\{0\}$ and C , so there exists $a \in \mathbb{R}^n \setminus \{0\}$ such that $0 \leq a^\top x$ for all $x \in C$. Every $x \in C$ is in the form $x = x_1 - x_2$, hence

$$0 \leq a^\top (x_1 - x_2) \iff a^\top x_2 \leq a^\top x_1.$$
■

Example 3. Let

$$\begin{aligned} C_1 &= \{(x, y) : y \geq e^x\} \\ C_2 &= \{(x, y) : y \leq 0\}. \end{aligned}$$

Then $C_1 \cap C_2 = \emptyset$ and $H = \{(x, y) : y = 0\}$.

PROPOSITION 4 (Strict separation theorem, book 1.5.3). *Let $C_1, C_2 \subseteq \mathbb{R}^n$ convex disjoint. Then C_1 and C_2 can be strictly separated if one of the following holds:*

1. $C_1 - C_2$ is closed;
2. C_1 closed and C_2 compact;
3. C_1, C_2 polyhedral;
4. C_1, C_2 closed and $R_{C_1} \cap R_{C_2} = L_{C_1} \cap L_{C_2}$;
5. C_1 is closed, C_2 polyhedral, $R_{C_1} \cap R_{C_2} \subseteq L_{C_1}$.

11.1 Hyperplanes

Proof. We only prove under condition 2): assume $C_1 = A$ closed and $C_2 = B$ compact. Define

$$d^* = \inf_{a \in A, b \in B} \|a - b\| \geq 0.$$

Consider a maximizing sequence (a_k, b_k) where $a_k \in A, b_k \in B$ such that $\|a_k - b_k\| \rightarrow d^*$. Since B is compact, $\{b_k\}_k$ is bounded, hence, up to subsequences, $b_k \rightarrow \bar{b} \in B$, since B is closed. This implies also that $\{a_k\}_k$ is bounded (otherwise $\|a_k - b_k\| \rightarrow +\infty$), so, up to subsequences, $a_k \rightarrow \bar{a} \in A$. We have shown that $\|a_k - b_k\| \rightarrow \|\bar{a} - \bar{b}\|$. But, observing that $\|a_k - b_k\| \rightarrow d^*$, by uniqueness of the limit, we have also that $d^* = \|\bar{a} - \bar{b}\|$. In particular $\bar{a}$ is the projection of $\bar{b}$ onto A , and $\bar{b}$ is the projection of $\bar{a}$ onto B . By projection theorem,

$$\begin{aligned} (\bar{b} - \bar{a})^\top \cdot (a - \bar{a}) &\leq 0 \quad \forall a \in A \\ (\bar{a} - \bar{b})^\top \cdot (b - \bar{b}) &\leq 0 \quad \forall b \in B. \end{aligned}$$

Consider $c = \bar{b} - \bar{a}$. Pick $\bar{x}$ on the segment connection $\bar{a}$ to $\bar{b}$, then

$$H = \bar{x} + \left\{ x : c^\top x = 0 \right\}$$

strictly separates A and B . ■

PROPOSITION 5 (Book 1.5.4). *The closure of the convex hull of a set C is the intersection of all the closed halfspaces containing C .*

Proof.

$$\begin{aligned} H &= \text{intersection of closed halfspaces containing } C \\ &= \bigcap \{ \tilde{H} \text{ s.t. } \tilde{H} \text{ is a closed halfspace, } C \subseteq \tilde{H} \}. \end{aligned}$$

We prove the double inclusion.

$\text{cl}(\text{conv}(C)) \subseteq H$: if $\tilde{H}$ is a closed halfspace containing C , then $\tilde{H}$ contains $\text{cl}(\text{conv}(C))$, so $\text{cl}(\text{conv}(C))$ is contained also in their intersection, which is H .

$H \subseteq \text{cl}(\text{conv}(C))$: we prove that $\mathbb{R}^n \setminus \text{cl}(\text{conv}(C)) \subseteq \mathbb{R}^n \setminus H$. Let $x \in \mathbb{R}^n \setminus \text{cl}(\text{conv}(C))$. Since $x \notin \text{cl}(\text{conv}(C))$, by strict separation point 2., we can separate $\{x\}$ from $\text{cl}(\text{conv}(C))$ via an hyperplane H^* , which is contained in H . The halfspace corresponding to H^* does not contain x , so $x \notin H$. Hence $H \subseteq \text{cl}(\text{conv}(C))$. ■

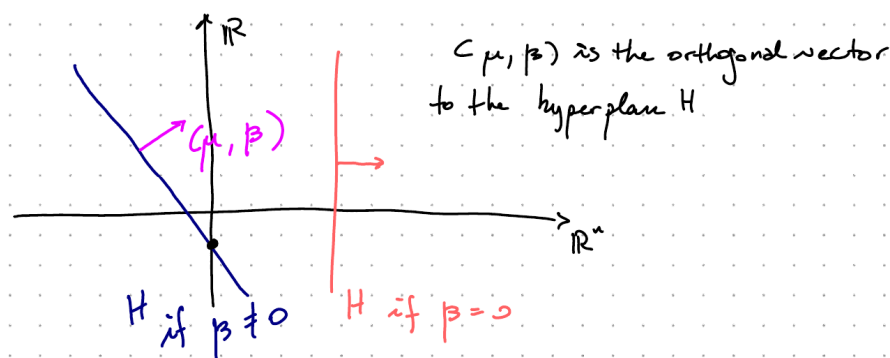
LECTURE 12 OF 08/04/2026

Let $C \subseteq \mathbb{R}^{n+1}$, $x \notin C$. Idea: if C does not contain vertical lines, we can strictly separate C from x with a "non-vertical" hyperplane.

For $(u, w) \in \mathbb{R}^{n+1}$, $u \in \mathbb{R}^n$, $w \in \mathbb{R}$. A hyperplane H is in the form

$$H = \{(u, w) : (u, w)^\top \cdot (\mu, \beta) = b\}$$

for $\mu \in \mathbb{R}^n$, $\beta \in \mathbb{R}$. We say that H is nonvertical if $\beta \neq 0$.



If $(\bar{u}, \bar{w}) \in H$, then $H = (\bar{u}, \bar{w}) + (u, w)$ such that $(u, w) \perp (\mu, \beta)$ (i.e. $(u, w)^\top \cdot (\mu, \beta) = 0$).

Which is the height at which H intersects the $(n+1)$ -axis? When the point $(0, \xi) \in H$? If $(0, \xi) = (\bar{u}, \bar{w}) + (u, w)$ for $(u, w) \perp (\mu, \beta)$. Multiply by (μ, β) , i.e. take the projection, so

$$\xi\beta = \bar{u}^\top \cdot \mu + \beta\bar{w} + 0 \implies \xi = \frac{\bar{u}^\top \cdot \mu}{\beta} + \bar{w}.$$

PROPOSITION 1 (Non-vertical hyperplane theorem, book 1.5.8). Let $C \subseteq \mathbb{R}^{n+1}$ nonempty convex set, assume C does not contain any vertical line. Then

1. C is contained in the closed halfspace corresponding to a non-vertical hyperplane, i.e there exists $\mu \in \mathbb{R}^n$, $0 \neq \beta \in \mathbb{R}$ and $\gamma \in \mathbb{R}$ such that

$$\mu^\top u + \beta w \geq \gamma \quad \forall (u, w) \in C;$$

2. if $(\bar{u}, \bar{w}) \notin \text{cl}(C)$, there exists a non-vertical hyperplane that strictly separates $(\bar{u}, \bar{w})$ and C .

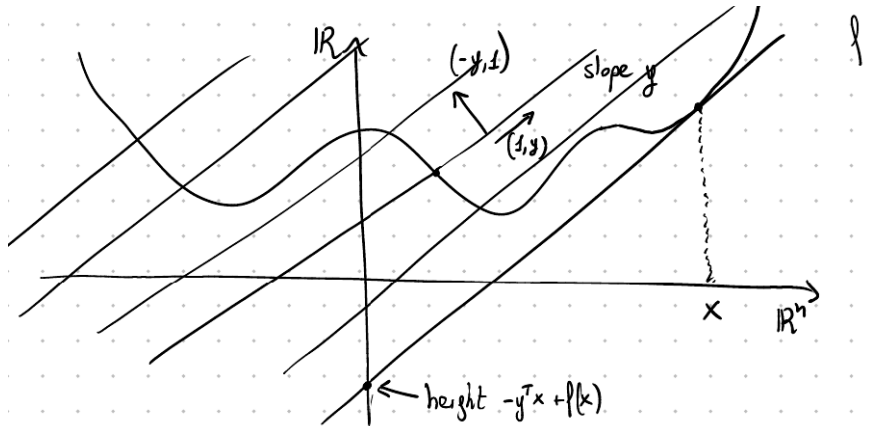
Proof. Not required. ■

12.1 CONVEX CONJUGATE FUNCTIONS

We would like to describe a function f as the supremum of a family of affine functions.

DEFINITION 1. Let $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$. We define the *conjugate* of f (or convex conjugate) as the function

$$f^*(y) = \sup_{x \in \mathbb{R}^n} y^\top x - f(x).$$



So what? Assume f given. We want to describe f as $\sup\{\text{affine function below } f\} = \sup\{\text{lines of any slope } y \in \mathbb{R}^n, \text{ lines below } f\}$.

Fix a slope $y \in \mathbb{R}^n$ and fix any $(x, f(x))$ in the graph of f . Then the line passing through $(x, f(x))$ and of slope y is

$$\{(x, f(x)) + (u, w) : (u, w) \perp (-y, 1)\}$$

where the "line" $(x, f(x)) + (u, w)$ intersects $(0, \zeta)$ at height $-y^\top x + f(x)$.

To keep f above the selected line of slope y , we need to use the one such that the quantity $-y^\top x + f(x)$ is infimized, so $\inf_{x \in \mathbb{R}^n} -y^\top x + f(x)$ is the intercept of the "highest" line with slope y that passes from a point in the graph of f and contains the graph of f in one of its closed halfspaces. Now we observe that

$$\inf_{x \in \mathbb{R}^n} -y^\top x + f(x) = \sup_{x \in \mathbb{R}^n} y^\top x - f(x) = -f^*(y).$$

DEFINITION 2 (Biconjugate function). The biconjugate function is defined as

$$f^{**}(x) = \sup_{y \in \mathbb{R}^n} x^\top y - f^*(y).$$

Example 3. Let $f(x) = x^2$. We want to compute its conjugate f^* .

From the definition, $f^*(y) = \sup_{x \in \mathbb{R}^n} \{yx - f(x)\}$. One strategy can be find the critical points of the function $g(x) = yx - x^2 \implies g'(x) = y - 2x \implies x = y/2$ is a critical point (in particular a maximum). Hence

$$f^*(y) = y \frac{y}{2} - \left(\frac{y}{2}\right)^2 = \frac{y^2}{4}.$$

Example 4. Let $f(x) = e^x$. Again, we can compute $g'(x)$ and find a critical point:

$$g(x) = yx - e^x \implies g'(x) = y - e^x \implies y = e^x \text{ critical point.}$$

Now, if $y > 0$ we are fine, since $x = \log(y)$. Otherwise? If $y < 0$ we would have $yx - e^x = \text{negative} \cdot x - e^x$ which explodes to $+\infty$ as $x \rightarrow -\infty$. If instead $y = 0$, $\sup_x -e^x = 0$. Finally,

$$f^*(y) = \begin{cases} y \log(y) - y & y > 0 \\ 0 & y = 0 \\ +\infty & y < 0 \end{cases}.$$

Example 5. Consider $f(x) = x - 1$. Then

$$f^*(y) = \sup_{x \in \mathbb{R}} yx - x + 1 = x(y - 1) + 1.$$

If $y \neq 1$, then we can get $x \rightarrow +\infty$ and we explode, hence

$$f^*(y) = \begin{cases} 1 & \text{if } y = 1 \\ +\infty & \text{otherwise.} \end{cases}$$

Example 6. Consider $f(x) = |x|$. Define

$$g(x) = \begin{cases} yx + x & \text{if } x \leq 0 \\ yx - x & \text{if } x \geq 0 \end{cases} = \begin{cases} x(y + 1) & \text{if } x \leq 0 \\ x(y - 1) & \text{if } x \geq 0 \end{cases}$$

We have 3 cases:

$$\begin{aligned} y + 1 < 0 &\implies x(y + 1) \nearrow +\infty \text{ as } x \searrow -\infty \\ y - 1 > 0 &\implies x(y - 1) \nearrow +\infty \text{ as } x \nearrow -\infty. \end{aligned}$$

12.1 Convex conjugate functions

If instead $|y| \leq 1$ we have

$$g(x) = \begin{cases} x \cdot \text{positive} & x \leq 0 \\ x \cdot \text{negative} & x \geq 0 \end{cases}$$

In both cases, the supremum is 0. Finally,

$$f^*(y) = \begin{cases} 0 & \text{if } |y| \leq 1 \\ +\infty & \text{otherwise} \end{cases} = \delta_{\{y: |y| \leq 1\}}$$

the convex indicator function of the set $\{y : |y| \leq 1\}$.

LECTURE 13 OF 14/04/2026

PROPOSITION 1 (Conjugate function theorem, Book 1.6.1). Let $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ and denote with f^* its conjugate, $f^{**} = (f^*)^*$. Then:

- a) $f(x) \geq f^{**}(x)$ for any $x \in \mathbb{R}^n$;
- b) if f is convex, then if one between f, f^*, f^{**} is proper, then also the other two are;
- c) if f is closed and convex, $f(x) = f^{**}(x)$ for every $x \in \mathbb{R}^n$;
- d) $f^* = (\text{cl}(f))^*$. Furthermore, if $\text{cl}(f)$ is proper, then $\text{cl}(f) = f^{**}$.

Proof. **a):** $f^*(y) = \sup_x y^\top x - f(x)$. This means $f^*(y) \geq y^\top x - f(x)$ for any y, x , this implies $y^\top x \leq f(x) + f^*(y)$ (this last is called Fenchel's inequality). Now,

$$f^{**}(x) = \sup_y x^\top y - f^*(y) \leq f(x).$$

b): assume f is convex and proper, so $\text{epi}(f)$ is nonempty convex set in $\mathbb{R}^{n+1}$, since f is proper, $\text{epi}(f)$ does not contain any vertical line, so there exists $(y, 1) \in \mathbb{R}^{n+1}$, $c \in \mathbb{R}$ such that $y^\top x + f(x) \geq c$ for any $x \in \mathbb{R}^n$ (because the epigraph lies above the tangent hyperplane). This implies $f^*(-y) \leq -c \implies \sup_x -y^\top x - f(x) \leq -c$, so $f^*(y) < +\infty$. Is $f^* \equiv -\infty$? No, indeed there exists $\bar{x}$ such that $f(\bar{x})$ is finite (f is proper), hence $f^*(y) = \sup_x y^\top x - f(x) \geq y^\top \bar{x} - f(\bar{x})$. This shows that f is proper (the same computation holds for the others).

c): from a) we know that $f^{**}(x) \leq f(x)$. We want to prove $f^{**}(x) \geq f(x)$, so $\text{epi}(f^{**}) \subseteq \text{epi}(f)$. By contradiction, assume $x \in \text{epi}(f^{**}) \setminus \text{epi}(f)$. Then we can separate a point $\{(x, \gamma)\}$ from $\text{epi}(f)$, also strictly, so there exists (y, ξ) , $\xi \neq 0$, $c \in \mathbb{R}$ such that

$$y^\top z + \xi w < c < y^\top x + \xi \gamma$$

for any $(z, w) \in \text{epi}(f)$. Observe that we can take $w \nearrow +\infty$, so we must have $\xi < 0$. Assume WLOG $\xi = -1$ (by scaling). So

$$y^\top z - f(z) < c < y^\top x - f^{**}(x) \quad \forall x \in \text{dom}(f).$$

Take the supremum over z , so

$$\begin{aligned}\sup_z f^*(y) &= \sup_z y^\top z - f(x) < y^\top - f^{**}(x) \\ \implies y^\top x &> f^*(y) + f^{**}(x)\end{aligned}$$

which contradicts the Fenchel's inequality.

d): define $\check{f}^* = (\check{\text{cl}}(f))^*$. Then $\check{\text{cl}}(f)$ is a convex closed function, so $\text{epi}(\check{\text{cl}}(f)) = \text{cl}(\text{conv}(\text{epi}(f)))$. We know $\text{cl}(\text{conv}(\text{epi}(f))) = \text{intersection of all the closed halfspaces that contains } \text{epi}(f)$. This implies that $-f^*(y)$ is intercept of the highest hyperplane such that $\text{epi}(f)$ is above that hyperplane, hence $-f^*(y) = \check{f}^*(y)$ because hyperplanes below f and $\check{\text{cl}}(f)$ are the same. This shows $f^* = \check{f}^*$, then $f^{**} = (\check{\text{cl}}(f))^* = \check{\text{cl}}(f)$ by point c). ■

13.1 CONVEX OPTIMIZATION

We move now the focus onto the actual optimization of convex functions. We set ourselves on the problem

$$\inf_{x \in X} f(x)$$

where $X \subseteq \mathbb{R}^n$, $f : X \rightarrow (-\infty, +\infty]$. We say that x^* is a minimum point if it is such that

$$x^* = \text{argmin}_{x \in X} f(x).$$

If $x \in X \cap \text{dom}(f)$, then $f(x^*) = \inf_{x \in X} f(x) = \min_{x \in X} f(x)$, so the infimum is attained as x^* .

We recall that we say that x^* is a

- local minima if $f(x^*) \leq f(x)$ for any $x \in B(x^*, \varepsilon)$ for some $\varepsilon > 0$;
- global minima if $f(x^*) \leq f(x)$ for any $x \in X$.

PROPOSITION 1 (Book 3.1.1). *Let $X \subseteq \mathbb{R}^n$ be convex, $f : X \rightarrow (-\infty, +\infty]$ be a convex function. Then local minima are global minima. In addition, if f is strictly convex, then there exists at most one global minima.*

Proof. Assume x^* local minima but not global, i.e. there exists $\bar{x}$ such that $f(\bar{x}) < f(x^*)$. Then

$$f(\alpha x^* + (1 - \alpha)\bar{x}) \leq \alpha f(x^*) + (1 - \alpha)f(\bar{x}) < f(x^*) \quad \forall \alpha \in [0, 1].$$

For $\alpha \rightarrow 1$, $\alpha x^* + (1 - \alpha)\bar{x} \in B(x^*, \varepsilon)$, where ε is such that x^* is local minima in $B(x^*, \varepsilon)$, which contradicts the fact that x^* is local minima, hence it is also a global one.

Furthermore, if f is strictly convex, assume x, x^* two global minima, so $f(x) = f(x^*)$. By strict convexity,

$$f\left(\frac{1}{2}x + \frac{1}{2}x^*\right) < \frac{1}{2}f(x) + \frac{1}{2}f(x^*) = f(x).$$

This shows that $x = x^*$. ■

The natural question now is: does a minimum point exists? We will see that under some condition this existence is guaranteed.

DEFINITION 2 (Coercivity). A function $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is *coercive* if for every sequence $\{x_k\}_k \subseteq \mathbb{R}^n$ such that $\|x_k\| \rightarrow +\infty$, we have

$$\lim_{k \rightarrow +\infty} f(x_k) \rightarrow +\infty.$$

PROPOSITION 3 (Weierstrass, Book 3.2.1). Let $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ closed proper and LSC. Assume one among the following:

1. $\text{dom}(f)$ is bounded;
2. there exists $\bar{\gamma}$ such that the set $\{x : f(x) \leq \bar{\gamma}\}$ is nonempty and bounded;
3. f is coercive.

Then there exists $x^* \in \mathbb{R}^n$ such that $f(x^*) \leq f(x)$ for any $x \in \mathbb{R}^n$, i.e. x^* is a global minimum point.

Proof. Let $\{x_k\}_k$ be a minimizing sequence for f , so $f(x_k) \rightarrow \inf_x f(x)$. Then $\{x_k\}_k \subseteq \text{dom}(f)$.

1. If 1., $\{x_k\}_k \subseteq$ bounded set;
2. if 2., $f(x_k) \leq \bar{\gamma}$ for k big enough, hence $\{x_k\}_k$ is bounded;
3. if 3., $\{x_k\}_k$ cannot "go to ∞ ", otherwise $f(x_k) \rightarrow +\infty$ since f is coercive.

13.1 Convex optimization

In any of these, the sequence $\{x_k\}_k$ is *bounded*. By BW, up to subsequences, $x_k \rightarrow \bar{x} \in \mathbb{R}^n$. Using the closeness and lower semicontinuity of f , it holds true that

$$f(\bar{x}) \leq \liminf_{k \rightarrow +\infty} f(x_k) = \inf_x f(x)$$

hence $\bar{x}$ is a global minimum point. ■

PROPOSITION 4 (Book 3.2.2). *Let $X \subseteq \mathbb{R}^n$ closed convex, $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ convex closed, assume $X \cap \text{dom}(f) \neq \emptyset$. Then the set of minima of f over X is nonempty if and only if X and f has no nonzero common direction of recession.*

Proof. Assume $\tilde{f} = \inf_{x \in X} f(x)$. Notice that $\tilde{f} < +\infty$, because $X \cap \text{dom}(f)$ is nonempty. Consider $\{\gamma_k\}_k$ such that $\gamma_k \rightarrow \tilde{f}$, $\gamma_k \geq \tilde{f}$ for any k . Define $V_k = \{x : f(x) \leq \gamma_k\}$. Then the minimum point belongs to the intersection $\bigcap_k (X \cap V_k)$. The problem now is to guarantee that the latter intersection is nonempty. In fact it is nonempty, indeed

$$R_X \cap R_{V_k} = R_X \cap R_f = \{0\}.$$

By theorem 1.4.2(c) of the book, this implies the intersection nonempty. ■

LECTURE 14 OF 15/04/2026

Example 1. Consider $f(x) = e^{-x}$, $X = (-\infty, 1]$. Then f is convex and closed. Then

$$\begin{aligned} r_f &= \{d : d \geq 0\} \\ R_X &= \{d : d \leq 0\}. \end{aligned}$$

Hence $r_f \cap R_X = \{0\}$, this implies that f attains the infimum over X .

PROPOSITION 2 (Book 3.2.4). *Let X be closed convex in $\mathbb{R}^n$, $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ convex and closed. Assume $X \cap \text{dom}(f)$ nonempty. Then the set of minima of f over X , denoted by X^* is nonempty under one among the following assumptions/conditions:*

1. $R_f \cap R_X = L_X \cap L_f$;
2. $R_X \cap R_f \subseteq L_f$ and X is polyhedral.

Furthermore, under 1., $X^* = \tilde{X} + (L_X \cap L_f)$ where $\tilde{X}$ is some compact set in $\mathbb{R}^n$.

Proof. Let $\tilde{f} = \inf_{x \in X} f(x) < +\infty$ as $X \cap \text{dom}(f)$ is nonempty. Let $\{\gamma_k\}_k$ such that $\gamma_k \rightarrow \tilde{f}$ and $\gamma_k \geq \tilde{f}$ for any k . Set $V_k = \{x : f(x) \leq \gamma_k\}$. Note $X^* = \bigcap_k (X \cap V_k)$.

Assume 1., then $\{X \cap V_k\}_k$ is a nestes sequence of closed convex sets. $R_X \cap V_k = R_X \cap R_{V_k} = R_X \cap R_f = L_X \cap L_f = L_X \cap L_{V_k} = L_{X \cap V_k}$. By 1.4.11(a) X^* is nonempty and it holds ist characterization.

Assume 2., then $R_X \cap R_{V_k} = R_X \cap R_f \subseteq L_f$, X polyhedral, so retractive. By 1.4.11(b), X^* is nonempty. ■

LECTURE 15 OF 21/04/2026

15.1 MINMAX PROBLEMS

Let $X, Y \subseteq \mathbb{R}^n$, $L : X \times Y \rightarrow \overline{\mathbb{R}}$. Define

$$f(x) = \sup_y L(x, y)$$

$$g(y) = \inf_x L(x, y).$$

So $f : X \rightarrow \mathbb{R}$, $g : Y \rightarrow \mathbb{R}$. We are interested in $\inf_x f(x)$ and $\sup_y g(y)$. How do they relate?

Remark 1. By definition, $g(y) \leq L(x, y) \leq f(x)$ for any $x \in X$ and $y \in Y$. In particular, $g(y) \leq f(x)$. Take the $\inf_x \sup_y$ on both sides, so

$$\sup_y g(y) \leq \inf_x f(x).$$

We get then $\sup_y \inf_x L(x, y) \leq \inf_x \sup_y L(x, y)$.

DEFINITION 2 (Saddle value). If $\sup_y \inf_x L(x, y) = \inf_x \sup_y L(x, y)$ we name this value *saddle value* of L .

DEFINITION 3 (Saddle point). A point $(\bar{x}, \bar{y})$ is a *saddle point* of L if

$$L(\bar{x}, y) \leq L(\bar{x}, \bar{y}) \leq L(x, \bar{y})$$

for any x, y .

THEOREM 4. A pair $(\bar{x}, \bar{y})$ is a saddle point if and only if $\bar{x}$ solves $\inf_x f(x)$, $\bar{y}$ solves $\sup_y g(y)$ and the saddle value exists, i.e. $\inf_x f(x) = \sup_y g(y)$.

Proof. " $\implies$ ": suppose $(\bar{x}, \bar{y})$ saddle point. Then $L(\bar{x}, y) \leq L(\bar{x}, \bar{y}) \leq L(x, \bar{y})$ for any x, y . This implies

$$\sup_y L(\bar{x}, y) \leq L(\bar{x}, \bar{y}) \leq \inf_x L(x, \bar{y}),$$

hence,

$$f(\bar{x}) \leq L(\bar{x}, \bar{y}) \leq g(\bar{y}) \leq f(\bar{x})$$

as $g \leq f$, so the inequalities become actually equalities.

The other implication is easy. ■

Why are we interested in this kind of problems? Our focus is $\inf_x f(x)$. Assume, by chance, that such an L exists and be such that $f(x) = \sup_y L(x, y)$. Now, how can we find Y and L ?

In this setting I can look at $\sup_y g(y)$ as a natural candidate of the "dual" problem, and hope that solving the dual is easier. We know for sure $g(y) \leq \inf_x f(x)$ for any y , hence $g(y)$ is a lower bound on the infimum in the primal problem.

Assume that exists $\bar{y}$ such that $g(\bar{y}) = \inf_x f(x)$. But also $L(x, \bar{y}) \leq f(x)$ for any x . So if $\bar{x}$ minimizes f , then $\bar{x}$ also minimizes $L(\cdot, \bar{y})$. So we can determine all minimizers of $L(\cdot, \bar{y})$ and search among them a minimizer of f . Observe that $g(\bar{y}) \leq \sup_y g(y) \leq \inf_x f(x) = g(\bar{y})$. Then $\bar{y}$ maximizes $g(y)$, hence $(\bar{x}, \bar{y})$ is a saddle point of L . But finding L is not easy...

But under convexity assumptions, all this process could be implemented easily. Let's focus on the convex case. The idea is to "embed" the $\inf_x f(x)$ into a "bigger" function F , so define $F : X \times U \rightarrow \overline{\mathbb{R}}$, where U is a set of perturbations, select F such that $F(x, 0) = f(x)$.

DEFINITION 5 (Optimal value function). The *optimal value function* $\varphi : U \rightarrow \overline{\mathbb{R}}$ is defined as

$$\varphi(u) = \inf_x F(x, u).$$

Observe that $\varphi(0) = \inf_x f(x)$.

THEOREM 6 (Book 3.3.1(a)). If F is convex (in both entries at the same time), then φ is convex.

Proof. If $\text{epi}(\varphi) = \emptyset$, then $\varphi \equiv +\infty$, which is convex. Otherwise, is $\text{epi}(\varphi)$ convex? Let $(\bar{u}, \bar{v}), (\tilde{u}, \tilde{v}) \in \text{epi}(\varphi)$. Then

$$\inf_x F(x, \bar{u}) = \varphi(\bar{u}) \leq \bar{v}$$

so there exists a sequence $\{\bar{x}_k\}_k \subseteq X$ such that $F(\bar{x}_k, \bar{u}) \rightarrow \varphi(\bar{u})$. We do the same reasoning for $\tilde{u}$, so there exists a sequence $\{\tilde{x}_k\}_k$ such that $F(\tilde{x}_k, \tilde{u}) \rightarrow \varphi(\tilde{u})$. Take $\alpha \in (0, 1)$, then

$$\varphi(\alpha\bar{u} + (1 - \alpha)\tilde{u}) \leq F(\alpha\bar{x}_k + (1 - \alpha)\tilde{x}_k, \alpha\bar{u} + (1 - \alpha)\tilde{u}).$$

Using the convexity of F (in both arguments), we get

$$\begin{array}{ccc} \alpha F(\bar{x}_k, \bar{u}) & + & (1 - \alpha)F(\tilde{x}_k, \tilde{u}) \\ \downarrow & & \downarrow \\ \varphi(\bar{u}) & & \varphi(\tilde{u}). \end{array}$$

Going to the limit,

$$\begin{aligned} \varphi(\alpha\bar{u} + (1 - \alpha)\tilde{u}) &\leq \alpha\varphi(\bar{u}) + (1 - \alpha)\varphi(\tilde{u}) \\ &\leq \alpha\bar{v} + (1 - \alpha)\tilde{v}. \end{aligned}$$

Therefore, $(\alpha\bar{u} + (1 - \alpha)\tilde{u}, \alpha\bar{v} + (1 - \alpha)\tilde{v}) \in \text{epi}(\varphi)$. ■

Example 7. Fix $f_0, f_1, \dots, f_m$ convex functions on C nonempty convex. Define

$$f(x) = \begin{cases} f_0(x) & \text{if } x \in C, f_i(x) \leq 0 \forall i = 1, \dots, m \\ +\infty & \text{otherwise} \end{cases}$$

Where we put the perturbations? The idea is to perturb the constraints, therefore

$$F(x, u) = \begin{cases} f_0(x) & \text{if } x \in C, f_i(x) \leq u_i \forall i = 1, \dots, m \\ +\infty & \text{otherwise} \end{cases}$$

Then $F : C \times U \rightarrow \overline{\mathbb{R}}$, and $f(x) = F(x, 0)$. Moreover, F is convex both in x and u .

Example 8. Same structure as above. Assume furthermore that

$$f_i(x) = h_i(A_i x + a_i) + \ell_i(x)$$

where $A_0, \dots, A_m$ are matrices, $h_i : Z_i \rightarrow \overline{\mathbb{R}}$, $a_i \in Z_i$ and ℓ_i are affine functions for every i .

Then $u = (u_1, \dots, u_m, z_1, \dots, z_m) \in \mathbb{R}^m \times Z_0 \times \dots \times Z_m =: U$. Then

$$F(x, u) = \begin{cases} h_0(A_0 x + a_0 - z_0) + \ell_0(x) & \text{if } x \in C, h_i(A_i x + a_i - z_i) + \ell_i(x) \leq u_i \\ +\infty & \text{otherwise} \end{cases}$$

Example 9. Consider the first example setting. Another possibility is to choose

$$F(x, u) = \begin{cases} f_0(x) + r\|u\|^2 & \text{if } x \in C, f_i(x) \leq u_i \\ +\infty & \text{otherwise} \end{cases}$$

DEFINITION 10 (Concave conjugate). Let g be concave. We define the concave conjugate as

$$g_*(y) = \inf_x y^\top x - g(x).$$

We define the *Lagrangian* L associated to the representation F the function

$$L(x, y) = \inf_u F(x, u) + y^\top u = (-F(x, \cdot))_*(y).$$

Assume $F(x, \cdot)$ closed convex for every fixed x . Then

$$L(x, y) = \left(\underbrace{F(x, \cdot)}_{\text{concave closed}} \right)_*(y) \implies (L(x, \cdot))_* = -F(x, \cdot).$$

Thus

$$\begin{aligned} -F(x, u) &= \inf_y y^\top u - L(x, y) \\ F(x, u) &= \sup_y L(x, y) - y^\top u \end{aligned}$$

if we assume $F(x, \cdot)$ convex closed.

Example 11. Set as the first example and consider $L(x, y) = \inf_u F(x, u) + y^\top u$. If $x \notin C$ we get $+\infty$. If some $y_i < 0$, then we can take u as large as we want and we get $-\infty$. Assume then $x \in C, y_i \geq 0$ for every i . Then

$$L(x, y) = \inf_{u: f_i(x) \leq u_i} f_0(x) + y^\top u.$$

We have to take the "smaller" possible u , thus $u_i = f_i(x)$. Under this condition, we get

$$L(x, y) = f_0(x) + \sum_{i=1}^m y_i f_i(x)$$

which is the standard Lagrangian in $\mathbb{R}^n$, therefore we are very happy.

LECTURE 16 OF 22/04/2026

THEOREM 1. *Book ???*

1. $L(x, \cdot)$ is closed and concave (the hypergraph below is closed and it is upper semi-continuous, i.e. $\{(y, w) : w \leq L(x, y)\}$);
2. if $F(x, \cdot)$ is closed convex for every fixed x , then $f(x) = \sup_y L(x, y)(\star)$;
3. if $L : X \times Y \rightarrow \overline{\mathbb{R}}$ is a function such that $(\star)$ holds, and if $L(x, \cdot)$ is closed concave for every x , then L is the Lagrangian associated with a uniquely determined representation $f(x) = F(x, 0)$ having F closed convex in u , with $F(x, u) = \sup_y L(x, y) - y^\top u$;
4. if $F(x, \cdot)$ is closed convex, then $L(\cdot, y)$ is convex for every fixed y if and only if F is convex in (x, u) .

Proof. **1.:** $L(x, y) = \inf_u F(x, u) + y^\top u = \inf_u$ affine function in y , which is closed and concave.

2.: we proved it last lecture, so

$$F(x, u) = \sup_y L(x, y) - y^\top u$$

but $f(x) = F(x, 0) = \sup_y L(x, y)$.

3.: follows from the definition.

4.: assume $F(x, \cdot)$ closed convex. Assume $L(\cdot, y)$ convex. Then the map $(x, u) \mapsto L(x, y) - y^\top u$ is convex in x for each fixed y . It is also convex in u for any y fixed. From 2., $F(x, u) = \sup_y L(x, y) - y^\top u$, which is convex as the supremum of a family of jointly convex functions in x and u .

Assume now F jointly convex in (x, u) . Then

$$L(x, y) = \inf_u F(x, u) + y^\top u$$

which, for any fixed y , is a convex function in (x, u) . Therefore, $L(\cdot, y)$ is convex as optimal value function.

From now on, we will denote with

(P) $\inf_x f(x)$ where $f(x) = \sup_y L(x, y)$ as the primal problem

(D) $\sup_y g(y)$ where $g(y) = \inf_x L(x, y)$ as the dual problem

and the optimal value function $\varphi(x) = \inf_u F(x, u)$.

THEOREM 2 (Main). 1. The function g in (D) is concave and closed. Indeed we have the following relation:

$$g = (-\varphi)_*, -g_* = \text{cl}(\text{co}(\varphi))$$

the convex closure of φ ;

2. if F is jointly convex in (x, u) then $-g_* = \text{cl}(\varphi)$ and, indeed,

$$\sup_y g(y) = \liminf_{u \rightarrow 0} \varphi(u)$$

assuming $0 \in \text{cl}(\text{dom}(\varphi))$.

Proof.

$$\begin{aligned} g(y) &= \inf_x L(x, y) \\ &= \inf_{x, u} F(x, u) + y^\top u \\ &= \inf_u \inf_x F(x, u) + y^\top u \\ &= \inf_u \varphi(u) + y^\top u \\ &= \inf_u y^\top u - (-\varphi)(u) = (-\varphi)_*(y). \end{aligned}$$

Furthermore,

$$g = (-\varphi)_* \implies g_* = (-\varphi)_{**}.$$

Observing that $-(-\varphi)_{**} = \varphi^{**}$ we conclude that $-g_* = \varphi^{**} = \check{\text{cl}}(\varphi)$.

If F is jointly convex in (x, u) we know that φ is convex, hence $\check{\text{cl}}(\varphi) = \text{cl}(\varphi)$. Since we know that the closure of φ is LSC, then $\sup_y g(y) = \inf_x f(x) = \text{cl}(\varphi)(0) = \liminf_{u \rightarrow 0} \varphi(u)$. ■

How if we reverse the problem? Consider $g(y) = G(y, 0)$, so that $G : Y \times V \rightarrow \overline{\mathbb{R}}$, $G(y, v) = \inf_x L(x, y) - v^\top x$. Define the *dual optimal value function* $\gamma(v) = \sup_y G(y, v)$. Observe that

$$G(y, v) = \inf_x \inf_u F(x, u) + y^\top u - v^\top x = -F^*(v, -y)$$

where

$$F^*(v, -y) = \sup_{x, u} (v, -y)^\top (x, u) - F(x, u).$$

THEOREM 3. 1. G concave closed, γ closed;

2. if F is convex (jointly in (x, u)) then

$$F(x, u) = \sup_{y, v} G(y, v) + x^\top v - y^\top u = -G^*(-x, u).$$

Moreover, $f = (-\gamma)^*$, $-f^* = \text{cl}(\gamma)$;

3. if f is convex, then

$$\inf_x f(x) = \limsup_{v \rightarrow 0} \gamma(v)$$

assuming $0 \in \text{cl}(\text{dom}(\gamma))$.

Exercise.

$$\begin{aligned} F(x, u) &= \sup_y L(x, y) - y^\top u \\ &= \sup_{y, v} G(y, v) + x^\top v - y^\top u \\ &= \sup_{y, v} -(-G(y, v)) - y^\top u - (-x)^\top v = -G^*(-x, u). \end{aligned}$$

Same argument as before for the rest. ■

Example 4. Consider the setting $f_0, f_1, \dots, f_m$, the objective is

$$\inf_{x \in C} f_0(x) \text{ such that } f_i(x) \leq 0.$$

We already computed $F(x, u)$ and $L(x, y)$. Then

$$g(y) = \inf_x L(x, y) = \begin{cases} \inf_x f_0(x) + \sum_i y_i f_i(x) & \text{if } y_i \geq 0 \\ -\infty & \text{otherwise} \end{cases}$$

Let's consider a subcase: the linear programming problem. We have

$$f_0(x) = \langle b_0, x \rangle + \beta_0, \quad b_0 \in \mathbb{R}^n, \quad \beta_0 \in \mathbb{R}$$

$$f_i(x) = \langle b_i, x \rangle + \beta_i, \quad b_i \in \mathbb{R}^n, \quad \beta_i \in \mathbb{R} \quad \forall i.$$

it is usually presented as

$$\inf c^\top x$$

$$Ax \leq b.$$

We seek for $g(y)$ such that $g(y) = G(y, 0)$. Who is $G(y, v)$? Recall that $L = f_0 + \sum_i y_i f_i$. Then

$$\begin{aligned} G(y, v) &= \inf_x L(x, y) - v^\top x \\ &= \inf_x \langle b_0, x \rangle + \beta_0 + \sum_{i=1}^m y_i [\langle b_i, x \rangle + \beta_i] - v^\top x \\ &= \inf_x \beta_0 + \sum_i y_i \beta_i + \langle x, b_0 + \sum_i y_i b_i - v \rangle \\ &= \begin{cases} \beta_0 + \sum_{i=1}^m y_i \beta_i & \text{if } v = b_0 + \sum_i y_i b_i \\ -\infty & \text{otherwise} \end{cases} \end{aligned}$$

We imposed that $b_0 + \sum_i y_i b_i - v = 0$ because if we took $x \searrow -\infty$, then $\langle x, b_0 + \sum_i y_i b_i - v \rangle \searrow -\infty$, therefore the infimum would be $-\infty$. Therefore the dual problem turns out to be

$$\begin{aligned} \sup_{y \geq 0} \quad & \beta_0 + \sum_{i=1}^m y_i \beta_i \\ & \beta_0 + \sum_{i=1}^m y_i b_i = 0. \end{aligned}$$

Remark 5. Observe that F jointly convex in (x, u) implies φ convex. Then also $\text{epi}(\varphi) = \text{projection of } \text{epi}(f) \text{ through the map } (x, u, \alpha) \mapsto (u, \alpha)$. Observe that

$\text{epi}(F)$ is polyhedral, indeed is the intersection of closed halfspaces, as each f_i is an affine function. Then $\text{epi}(\varphi)$ is a linear map applied to a polyhedral set. This shows that $\text{epi}(\varphi)$ is closed (as seen in the previous lecture).

LECTURE 17 OF 28/04/2026

17.1 FENCHEL'S DUALITY

Consider

$$\begin{aligned} h : \mathbb{R}^n &\rightarrow \overline{\mathbb{R}} \text{ proper convex} \\ K : \mathbb{R}^m &\rightarrow \overline{\mathbb{R}} \text{ proper convex} \\ A : \mathbb{R}^n &\rightarrow \mathbb{R}^m \text{ linear application} \end{aligned}$$

and set the problem to be

$$\inf_x f(x), \quad f(x) = h(x) + K(Ax).$$

Then the perturbation function is

$$F(x, u) = h(x) + K(Ax + u).$$

The Lagrangian is

$$\begin{aligned} L(x, y) &= \inf_u F(x, u) + y^\top u \\ &= \inf_u h(x) + K(Ax + u) + y^\top u \\ &= h(x) + \inf_u K(Ax + u) + y^\top u \quad \tilde{u} = Ax + u \\ &= h(x) + \inf_{\tilde{u}} K(\tilde{u}) + y^\top (\tilde{u} - Ax) \\ &= h(x) - \sup_{\tilde{u}} (-y)^\top \tilde{u} - K(\tilde{u}) - y^\top Ax \\ &= h(x) - K^*(-y) - y^\top Ax \end{aligned}$$

assuming $h(x) < +\infty$.

In the dual, consider the perturbation function

$$\begin{aligned}
G(y, v) &= -F^*(v, -y) \\
&= -\sup_{(x, u)} v^\top x - y^\top u - h(x) - K(Ax + w) \\
&= -\sup_{(x, \tilde{u})} v^\top x - y^\top \tilde{u} + y^\top Ax - h(x) - K(\tilde{u}) \\
&= -\sup_x \underbrace{v^\top x + y^\top Ax}_{=(v + A^\top y)^\top x} - h(x) - \sup_{\tilde{u}} (-y)^\top \tilde{u} - K(\tilde{u}) \\
&= -h^*(v + A^\top y) - K^*(-y).
\end{aligned}$$

So $g(y) = G(y, 0) = -h^*(A^\top y) - K^*(-y)$, hence the dual problem is

$$\sup_y -h^*(A^\top y) - K^*(-y).$$

17.2 SUBDIFFERENTIALS

Recall 1. If $\varphi : U \rightarrow \overline{\mathbb{R}}$ is convex, then φ is continuous on $\text{int}(\text{dom}(f))$

Recall 2. $0 \in \text{int}(\text{dom}(f)) \iff \{y : \varphi^*(y) \leq \beta\}$ is compact for any β and one subset of this type is nonempty, $\varphi(0) < +\infty$.

Recall 3. Let φ convex. If one of the sublevel sets is compact, then all sublevel sets are compact.

Recall 4. For any φ , the convex conjugate function $\varphi^*(y) = \sup_u y^\top u - \varphi(u)$ is convex.

What about differentiability? In general, convexity does not imply differentiability, e.g. $\varphi(x) = |x|$.

DEFINITION 1 (Subgradient/subdifferential). Let $\varphi : U \rightarrow \overline{\mathbb{R}}$, $u \in U$. We say that $y \in U$ is a *subgradient* of φ at u if

$$\varphi(v) \geq \varphi(u) + \langle y, v - u \rangle$$

for any $v \in U$. Furthermore denote as the *subdifferential* of φ at u the set

$$\partial\varphi(u) = \{y : y \text{ subgradient of } \varphi \text{ at } u\}.$$

Remark 2. If φ is differentiable at u , then $\partial\varphi(u) = \{\nabla\varphi(u)\}$.

Assume now $\varphi : U \rightarrow \overline{\mathbb{R}}$ convex proper, $\varphi \neq +\infty$.

closed and convex: consider $y_1, y_2 \in \partial\varphi(u)$, $\alpha \in (0, 1)$. Then

- $\partial\varphi(u)$ closed and convex:

$$\begin{aligned}\varphi(v) &\geq \varphi(u) + \langle y_1, v - u \rangle \forall v \\ \varphi(v) &\geq \varphi(u) + \langle y_2, v - u \rangle \forall v.\end{aligned}$$

Multiply the first by α and second by $(1 - \alpha)$. Then

$$\begin{aligned}\alpha\varphi(v) + (1 - \alpha)\varphi(v) &\geq \alpha\varphi(u) + (1 - \alpha)\varphi(u) + \langle \alpha y_1 + (1 - \alpha)y_2, v - u \rangle \\ \varphi(v) &\geq \varphi(u) + \langle \alpha y_1 + (1 - \alpha)y_2, v - u \rangle \forall v.\end{aligned}$$

Therefore, $\alpha y_1 + (1 - \alpha)y_2 \in \partial\varphi(u)$. Consider now a sequence $\{y_k\}_k \subseteq \partial\varphi(u)$ such that $y_k \rightarrow \bar{y}$. Then

$$\begin{aligned}\varphi(v) &\geq \varphi(u) + \langle y_k, v - u \rangle \\ &\quad \downarrow \\ \varphi(v) &\geq \varphi(u) + \langle \bar{y}, v - u \rangle\end{aligned}$$

Hence $\bar{y} \in \partial\varphi(u)$.

- If φ is continuous at u , then $\partial\varphi(u)$ is bounded: let $y \in \partial\varphi(u)$, then $\varphi(v) \geq \varphi(u) + \langle y, v - u \rangle$. By continuity of φ in u , there exists $\delta > 0$ such that $|\varphi(v) - \varphi(u)| \leq 1$ for any $v \in B(u, \delta)$. Then

$$\langle y, v - u \rangle \leq |\varphi(v) - \varphi(u)| \leq 1 \implies \sup_{\|z\| \leq \delta} \langle y, z \rangle \leq 1.$$

We know that $\|y\| = \sup_{\|x\| \leq 1} \langle x, y \rangle$, therefore

$$\delta\|y\| \leq 1 \implies \|y\| \leq 1/\delta$$

showing that $\partial\varphi(u)$ is bounded.

- $\varphi(\bar{u}) = \inf_u \varphi(u) \iff 0 \in \partial\varphi(\bar{u})$:

$$\begin{aligned}0 \in \partial\varphi(\bar{u}) &\implies \varphi(v) \geq \varphi(\bar{u}) + \langle 0, v - \bar{u} \rangle \implies \varphi(v) \geq \varphi(\bar{u}) \implies \bar{u} \text{ is a min} \\ \varphi(\bar{u}) = \inf_u \varphi(u) &\implies \varphi(v) \geq \varphi(\bar{u}) \implies \varphi(v) \geq \varphi(\bar{u}) + \langle 0, v - \bar{u} \rangle\end{aligned}$$

- if $\partial\varphi(u) \neq \emptyset$, then $\varphi(u) = \varphi^{**}(u)$: let $y \in \partial\varphi(u)$. Then $\varphi(v) \geq \varphi(u) + \langle y, v - u \rangle =$ convex closed function below φ . We know φ^{**} is the greatest convex closed function below φ , hence $\varphi(v) \geq \varphi^{**}(v) \geq \varphi(u) + \langle y, v - u \rangle$. If $u = v$, then we have equality.

- $y \in \partial\varphi(u) \iff \varphi(u) + \varphi^*(u) = \langle y, u \rangle$. In this case $y \in \text{dom}(\varphi^*)$.

" $\implies$ " $y \in \partial\varphi(u)$

$$\begin{aligned}\varphi(v) &\geq \varphi(u) + \langle y, v - u \rangle \\ \langle y, u \rangle - \varphi(u) &\geq \langle y, v \rangle - \varphi(v) \\ \langle y, u \rangle - \varphi(u) &\geq \sup_v \langle y, v \rangle - \varphi(v) \\ \langle y, u \rangle - \varphi(u) &\geq \varphi^*(y)\end{aligned}$$

Therefore, by Fenchel's inequality, we get the claim.

" $\Leftarrow$ " reverse the last argument.

- if $\varphi(u) = \varphi^{**}(u)$ then $\partial\varphi(u) = \partial\varphi^{**}(u)$: if $y \in \partial\varphi^{**}(u)$, then $\varphi^{**}(v) \geq \varphi^{**}(u) + \langle y, v - u \rangle \implies \varphi^{**}(v) \geq \varphi(u) + \langle y, v - u \rangle$. We know $\varphi(v) \geq \varphi^{**}(v)$ for every v , hence $y \in \partial\varphi(u)$, therefore $\partial\varphi^{**}(u) \subseteq \partial\varphi(u)$. If instead $y \in \partial\varphi(u)$, then $\varphi(u) < \infty$ and $\varphi(v) \geq \varphi(u) + \langle y, v - u \rangle = \varphi^{**}(u) + \langle y, v - u \rangle$, which the last is a closed convex function below φ , but we know that φ^{**} is the biggest of such a function, therefore $\varphi^{**}(v) \geq \varphi^{**}(u) + \langle y, v - u \rangle$, hence $y \in \partial\varphi^{**}(u)$.
- $y \in \partial\varphi(u) \iff u \in \partial\varphi^*(y)$ and $\varphi(u) = \varphi^{**}(u)$: let $y \in \partial\varphi(u)$, by the above point, $\varphi(u) + \varphi^*(u) = \langle y, u \rangle \implies \varphi(u) \geq \varphi^{**}(u) = \sup_z \langle z, u \rangle - \varphi^*(u) \geq \langle y, u \rangle - \varphi^*(y) = \varphi(u) \implies \varphi(u) = \varphi^{**}(u)$. Recall that $\varphi^*(z) = \sup_v z^\top v - \varphi(v) \geq z^\top u - \varphi(u) = z^\top u - y^\top u + \varphi^*(y)$, hence $\varphi^*(z) \geq \varphi^*(y) + \langle u, z - y \rangle$, therefore $u \in \partial\varphi^*(y)$.

COROLLARY 3. If $\varphi = \varphi^{**}$, then $y \in \partial\varphi(u) \iff u \in \partial\varphi^*(y)$ for every y .

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If $\varphi(0) = \varphi^{**}(0)$ then $\partial\varphi(0) = \{y : y \text{ is a minimizer of } \varphi^*\}$.

Comment: Let φ convex and continuous on $\text{int}(\text{dom}(\varphi))$. Previously we said φ restricted to $\text{ri}(\text{dom}(f))$ is continuous. Consider $\varphi : \mathbb{R}^2 \rightarrow \overline{\mathbb{R}}$,

$$\varphi(x, y) = \begin{cases} 0 & \text{if } 0 \leq x \leq 1, y = 0 \\ +\infty & \text{otherwise} \end{cases}$$

note that $\text{dom}(f) = \{(x, y) : x \in [0, 1], y = 0\}$, so a segment, therefore $\text{int}(\text{dom}(f)) = \emptyset$. But $\text{ri}(\text{dom}(f)) = \{(x, y) : x \in (0, 1), y = 0\} =: A$. Then $\varphi : A \rightarrow \overline{\mathbb{R}}$ is continuous.

18.1 PRIMAL/DUAL SETTING

Let $f : X \rightarrow \overline{\mathbb{R}}$, $F : X \times U \rightarrow \overline{\mathbb{R}}$ such that $f(x) = F(x, 0)$, $\varphi : U \rightarrow \overline{\mathbb{R}}$, $\varphi(u) = \inf_x F(x, u)$, $L : X \times Y \rightarrow \overline{\mathbb{R}}$, $g : Y \rightarrow \overline{\mathbb{R}}$, $G : Y \times V \rightarrow \overline{\mathbb{R}}$ such that $g(y) = G(y, 0)$, $\gamma : V \rightarrow \overline{\mathbb{R}}$, $\gamma(v) = \sup_y G(y, v)$.

Denote as

$$(P) \text{ minimize } f \text{ over } X, \inf_x f(x) = \inf(P)$$

$$(D) \text{ maximize } g \text{ over } Y, \sup_y g(y) = \sup(D)$$

Note that $\bar{x}$ solves $(P) \iff 0 \in \partial f(\bar{x})$ and $\bar{y}$ solves (D) if $0 \in \partial g(\bar{y})$, which the last one is a notational horror as, if g is concave, $\partial g(y)$ denotes the *superdifferential* of g at y , i.e. $z \in \partial g(y)$ if $g(w) \leq g(y) + \langle z, w - y \rangle$.

We now look at L : we assert that a point $(v, u) \in \partial L(\bar{x}, \bar{y})$ if $v \in \partial_x L(\bar{x}, \bar{y})$ and $u \in \partial_y L(\bar{x}, \bar{y})$, which means that

$$\begin{aligned} L(x, \bar{y}) &\geq L(\bar{x}, \bar{y}) + \langle v, x - \bar{x} \rangle \quad \forall x \\ L(\bar{x}, y) &\leq L(\bar{x}, \bar{y}) + \langle u, y - \bar{y} \rangle \quad \forall y. \end{aligned}$$

Notation: we dub the relation $(0, 0) \in \partial L(\bar{x}, \bar{y})$ as the *Khun-Tucker condition* for (P) .

THEOREM 1 (Book 15). *The implications $(a) \iff (b) \implies (c) \implies (d)$ hold among the following conditions:*

- (a) $\inf(P) = \sup(D)$;
- (b) $\varphi(0) = \text{cl}(\text{conv}(\varphi))(0)$;
- (c) *the saddle value of L exists*;
- (d) $\gamma(0) = \text{cl}(\gamma)(0)$.

Assuming $F(x, \cdot)$ closed convex for all x , then $(a) \iff (b) \iff (c)$. Assuming F jointly convex, they are all equivalent. Furthermore, it holds also $(e) \implies (f)$ where

- (e) $\bar{x}$ solves (P) , $\bar{y}$ solves (D) , $\inf(P) = \sup(D)$
- (f) $(\bar{x}, \bar{y})$ satisfies the Khun-Tucker condition.

Assuming $F(x, \cdot)$ closed convex for every x , then $(e) \iff (f)$.

Proof. Not required. ■

COROLLARY 2. *Suppose $\inf(P) = \sup(D)$ and $\sup(D)$ is attained. then $\bar{x}$ solves $(P) \iff \exists \bar{y}$ such that $(\bar{x}, \bar{y})$ satisfies the Khun-Tucker condition.*

THEOREM 3. *Let $\bar{y} \in Y$, then the following are equivalent:*

- (a) $\bar{y}$ solves (D) and $\inf(P) = \sup(D)$;
- (b) $-\bar{y} \in \partial\varphi(0)$;
- (c) $\inf_x L(x, \bar{y}) = \inf_x f(x)$.

Proof. We know that $-g(y) = \varphi^*(-y)$ and $\sup(D) = \varphi^{**}(0)$. Condition (a) is equivalent to $0 \in \partial\varphi^*(-y)$ because $0 \in \partial g(\bar{y})$, hence $\varphi^{**}(0) = \varphi(0)$, then apply the first proposition of today, so it follows $(a) \iff (b)$.

To prove $(a) \iff (c)$ observe that $g(\bar{y}) = \inf(P)$ is a rewrite of (c) and use the inequality $g(y) \leq \inf(P)$. ■

THEOREM 4. *If F is jointly convex, then φ is convex. Assume φ is bounded on a neighbourhood of 0, then $\inf(P) = \sup(D)$ and there exists at least one $\bar{y}$ such that $\bar{y}$ solves (D) .*

Proof. Under our assumptions, φ is continuous at 0, and everything follows. ■

Remark 5. We are happy also if $0 \in \text{int}(\text{dom}(\varphi))$.

Example 6. Consider the problem

$$\inf_x f_0(x) \quad \text{s.t. } f_i(x) \leq 0.$$

Then

$$\varphi(u) = \inf_x \{f_0(x) : x \in C, f_i(x) \leq u_i\}.$$

Observe that $\text{dom}(\varphi) = \{u \in \mathbb{R}^m : \exists x \in C \text{ s.t. } f_i(x) \leq u_i\}$. Therefore $0 \in \text{dom}(\varphi)$ if there exists x such that $f_i(x) < 0$ for every i .

We will now prove strong duality for the Fenchel's problem. In this setting we have $\varphi(u) = \inf_x h(x) + K(Ax + u)$, and the F that we defined is indeed jointly convex. To have $0 \in \text{dom}(\varphi)$ we can ask that there exists $\bar{x} \in \text{dom}(h)$ such that K is bounded from above in a neighbourhood of $A\bar{x}$.

What about subdifferentials? $(0, 0) \in \partial L(\bar{x}, \bar{y})$ if $\bar{x}$ solves $\inf(P)$ and $\bar{y}$ solves $\sup(D)$. The condition is then $L(x, \bar{y}) \geq L(\bar{x}, \bar{y})$ for any x . Recall that $L(x, y) = h(x) - K^*(-y) - y^\top Ax$. Therefore,

$$\begin{aligned} h(x) - K^*(-y) - y^\top Ax &\geq h(\bar{x}) - K^*(-y) - y^\top A\bar{x} \\ h(x) &\geq h(\bar{x}) + \langle A^\top \bar{y}, x - \bar{x} \rangle \quad \forall x. \end{aligned}$$

This condition is equivalent to $A^\top y \in \partial h(\bar{x})$. With a similar argument for the second case we get $A\bar{x} \in \partial K^*(-\bar{y})$.

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Example 1 (Quadratic programming). Consider the problem

$$\inf_x \frac{1}{2} x^\top Q x + c^\top x \text{ s.t. } Ax \leq b$$

where $Q \in \mathbb{R}^{n \times n}$ SPD, $c \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$.

The perturbation function is

$$F(x, u) = \begin{cases} \frac{1}{2} x^\top Q x - c^\top x & \text{if } Ax - b \leq u \\ +\infty & \text{otherwise} \end{cases}$$

The Lagrangian turns out to be

$$\begin{aligned} L(x, y) &= \inf_u F(x, u) + y^\top u \\ &= \inf_{u \in \mathbb{R}^m, Ax - b \leq u} \frac{1}{2} x^\top Q x - c^\top x + y^\top u. \end{aligned}$$

Observe that if there exists an index k such that $y_k \leq 0$ then we can pick u to be $u = (M, \dots, M, \underbrace{\lambda}_{\text{position } k}, M, \dots, M)$ and take $\lambda \rightarrow -\infty$, so $y_k \lambda \searrow -\infty$. Hence

we want $y \geq 0$. Then we have to take the smallest possible entries for u that are given by $Ax - b$. Finally,

$$L(x, y) = \begin{cases} \frac{1}{2} x^\top Q x - c^\top x + y^\top (Ax - b) & \text{if } y \geq 0 \\ -\infty & \text{otherwise.} \end{cases}$$

Now we derive the dual function g :

$$\begin{aligned} g(y) &= \inf_x L(x, y) \\ &= \inf_{x \in \mathbb{R}^n} \frac{1}{2} x^\top Q x - c^\top x + y^\top (Ax - b) \text{ assuming } y \geq 0 \end{aligned}$$

Consider

$$\frac{\partial L}{\partial x}(x, y) = \nabla_x L(x, y) = Qx - c + A^\top y$$

therefore

$$\nabla_x L(x, y) = 0 \implies Qx + c + A^\top y = 0 \implies x = Q^{-1}(-c - A^\top y).$$

Notice that we are allowed to invert Q as it is SPD, therefore the inverse exists. We now substitute in L :

$$L(Q^{-1}(-c - A^\top y), y) = -\frac{1}{2} (c + A^\top y)^\top Q^{-1} (c + A^\top y) - b^\top y$$

assuming $y \geq 0$. Finally, the dual problem is

$$\sup_{y \geq 0} -\frac{1}{2} (c + A^\top y)^\top Q^{-1} (c + A^\top y) - b^\top y.$$

Do we have strong duality? Yes, observe that F is jointly convex, and it is bounded in a neighbourhood of $u = 0$.

Example 2 (Second order cone programming). Consider the problem

$$\inf_{x \in \mathbb{R}^n} f^\top x$$

under the constraints $\|Ax + b\| \leq c^\top x + d$, where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, $d \in \mathbb{R}$.

The perturbation function is

$$F(x, (u, z)) = \begin{cases} f^\top x & \text{if } \|Ax + b - z\| - c^\top x - d \leq u \\ +\infty & \text{otherwise} \end{cases}$$

with $z \in \mathbb{R}^m$ and $u \in \mathbb{R}$.

The Lagrangian is then

$$L(x, (y, w)) = \inf_{(u, z): \|Ax - b - z\| - c^\top x - d \leq u} f^\top x + yu + w^\top z$$

where we call (y, w) the "dual variable", $y \in \mathbb{R}$, $w \in \mathbb{R}^m$. Again, if $y < 0$ then we get $-\infty$. We take then u as small as possible, so $u = \|Ax + b - z\| - c^\top x - d$. Therefore

$$\begin{aligned}
L(x, (y, w)) &= \inf_z f^\top x + y \left(\|Ax + b - z\| - c^\top - d \right) + w^\top z \\
&= f^\top x - yc^\top x - yd + \inf_z w^\top z + y\|Ax + b - z\| \quad \tilde{z} = Ax + b - z \\
&= f^\top x - yc^\top x - yd + \inf_{\tilde{z}} w^\top (Ax + b - \tilde{z}) + y\|\tilde{z}\| \\
&= f^\top x - yc^\top x - yd + w^\top (Ax + b) + \inf_{\tilde{z}} -w^\top \tilde{z} + y\|\tilde{z}\| \\
&= f^\top x - yc^\top x - yd + w^\top (Ax + b) - \sup_{\tilde{z}} w^\top \tilde{z} - y\|\tilde{z}\| \\
&= f^\top x - yc^\top x - yd + w^\top (Ax + b) - y \left(\sup_z \left(\frac{w^\top}{y} \right) z - \|z\| \right).
\end{aligned}$$

In parenthesis we have the conjugate of the norm function $(\|\cdot\|)^* \left(\frac{w}{y} \right)$. We recall that the convex conjugate of the norm function is the convex indicator function of the unit ball. Indeed $(\|\cdot\|)(v) = \sup_{z \in \mathbb{R}^n} v^\top z - \|z\|$. Recall that, in ℓ^2 it holds that

$$\|x\| = \sup_{\|y\| \leq 1} x^\top y.$$

Therefore,

$$\begin{aligned}
(\|\cdot\|)^*(v) &= \sup_{z \in \mathbb{R}^n} v^\top z - \|z\| \\
&= \sup_{\|z\|=1, \lambda \geq 0} \lambda v^\top z - \lambda \\
&= \sup_{\lambda \geq 0} \lambda \left(\sup_{\|z\|=1} v^\top z - 1 \right) \\
&= \sup_{\lambda \geq 0} \lambda (\|v\| - 1) \\
&= \begin{cases} 0 & \text{if } \|v\| \leq 1 \\ +\infty & \text{otherwise} \end{cases} = \delta_{\{(y,w): \|w/y\| \leq 1\}}.
\end{aligned}$$

In the end,

$$L(x, (y, w)) = f^\top x - yc^\top x - yd + w^\top (Ax + b) + = \delta_{\{(y,w): \|w/y\| \leq 1\}}.$$

The dual function is then

$$\begin{aligned}
 g((y, w)) &= \inf_x L(x, (y, w)) \\
 &= \inf_x \left(f - yc + A^\top w \right)^\top - yd + w^\top b = \delta_{\{(y, w): \|w/y\| \leq 1\}} \\
 &= \begin{cases} -yd + w^\top b = \delta_{\{(y, w): \|w/y\| \leq 1\}} & \text{if } f - yc + A^\top w = 0 \\ -\infty & \text{otherwise.} \end{cases}
 \end{aligned}$$

The dual problem is then

$$\sup_{(y, w)} -yd + w^\top b$$

under the constraints $f - yc + A^\top w = 0, \|w\| \leq |y|$.

LECTURE 20 OF 06/05/2026

20.1 PROXIMAL MAPPINGS

The topic is now the *proximal mappings*, we will consider the book Convex Analysis by Dirk Lorentz and we will see proximal algorithms and proximal mappings (chapter 8 + piece 10 + piece 12).

DEFINITION 1 (Proximal mapping/Monreau envelope). Let $f : \mathbb{R}^d \rightarrow \overline{\mathbb{R}}$ proper convex closed, $\lambda \geq 0$. We define the *proximal mapping* the map $\text{prox}_{\lambda f} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ defined as

$$\text{prox}_{\lambda f}(x) = \underset{y \in \mathbb{R}^d}{\text{argmin}} \lambda f(y) + \frac{1}{2} \|y - x\|^2.$$

We can think it as a "minimizer of the rescaled function λf + quadratic term". We define the *Monreau envelope* as

$$M_{\lambda f}(x) = \inf_{y \in \mathbb{R}^d} f(y) + \frac{1}{2\lambda} \|y - x\|^2.$$

THEOREM 2. Let $f : \mathbb{R}^d \rightarrow \overline{\mathbb{R}}$ be proper convex closed and fix $\lambda > 0$. The following holds:

1. for every $x \in \mathbb{R}^d$ there exists a unique $\hat{x}$ such that $\hat{x} = \text{prox}_{\lambda f}(x)$;
2. $\hat{x}$ as above is characterized by the variational inequality

$$\langle x - \hat{x}, y - \hat{x} \rangle + \lambda f(\hat{x}) - f(y) \leq 0$$

for every $y \in \mathbb{R}^d$;

3. x^* is a minimizer of f if and only if x^* is a fixed point of $\text{prox}_{\lambda f}$, i.e. $x^* = \text{prox}_{\lambda f}(x^*)$;
4. the Monreau envelope $M_{\lambda f}$ is a convex differentiable function and $\nabla M_{\lambda f} = \frac{1}{\lambda} (x - \text{prox}_{\lambda f}(x))$;

5. it holds that

$$\operatorname{argmin}_x f(x) = \operatorname{argmin}_x M_{\lambda f}(x).$$

Proof. It's a bluff. ■

Example 3. Let C be nonempty convex closed, $f(x) = \delta_C(x)$ the convex indicator function. Then

$$\begin{aligned} \operatorname{prox}_{\lambda f}(x) &= \operatorname{argmin}_{y \in \mathbb{R}^d} \lambda \delta_C(y) + \frac{1}{2} \|y - x\|^2 \\ &= \operatorname{argmin}_{y \in C} \frac{1}{2} \|y - x\|^2 = P_C(x) \end{aligned}$$

the projection function of C . Moreover,

$$\begin{aligned} M_{\lambda f} &= \frac{1}{2\lambda} \|P_C(x) - x\|^2 \\ &= \frac{1}{2\lambda} d(x, C)^2 \end{aligned}$$

where $d(x, C) = \inf_{y \in C} \|x - y\|$.

Example 4. Let $f(x) = |x|$, so $\hat{x} = \operatorname{prox}_{\lambda f}(x) = \operatorname{argmin}_y \lambda |y| + \frac{1}{2}(y - x)^2$.

Observe that if $x > 0$ then $\hat{x} > 0$ and $\operatorname{prox}_{\lambda f}(-x) = -\operatorname{prox}_{[\lambda f]}(x)$, so we can compute only for $x \geq 0$, with $y \geq 0$. The minimum of the objective function is achieved when its derivative is 0 (notice that the "inside function" is differentiable), hence $\lambda + y - x = 0 \implies y = x - \lambda$.

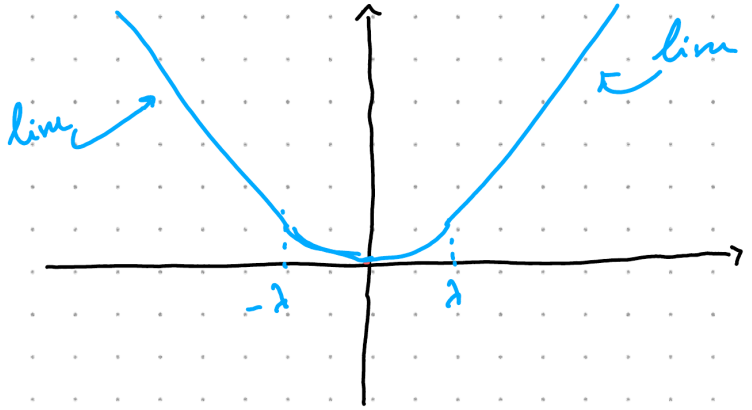
All in all,

$$\operatorname{prox}_{\lambda|\cdot|}(x) = \begin{cases} x - \lambda & \text{if } x \geq \lambda \\ 0 & \text{if } -\lambda < x < \lambda \\ x + \lambda & \text{if } x \leq -\lambda \end{cases} = \max(|x| - \lambda, 0) \operatorname{sgn}(x).$$

The Moreau envelope is the

$$M_{\lambda f}(x) = \begin{cases} |x| - \frac{\lambda}{2} & \text{if } |x| \geq \lambda \\ \frac{1}{2\lambda} x^2 & \text{if } |x| < \lambda \end{cases}$$

which is called *Huber function*, a differentiable approximation of $|\cdot|$.


 Figure 20.1: Monreau envelope of $|\cdot|$ function

PROPOSITION 5. 1. If $g(x) = \tau f(\mu x)$, $\tau, \mu \in \mathbb{R}$, $\tau > 0$ then

$$\text{prox}_{\lambda g}(x) = \frac{1}{\mu} \text{prox}_{\lambda \mu^2 \tau f}(x);$$

2. if $g(x) = f(Qx)$ with Q orthonormal matrix, then

$$\text{prox}_{\lambda g}(x) = Q^\top \text{prox}_{\lambda f}(Qx).$$

20.1.1 Monreau identity

THEOREM 6 (Monreau identity). Let $f : \mathbb{R}^d \rightarrow \overline{\mathbb{R}}$ proper closed convex, $f^*(y) = \sup_x y^\top x - f(x)$. Then

$$x = \text{prox}_f(x) + \text{prox}_{f^*}(x).$$

To prove this we need the following:

LEMMA 7. Let $f, g : \mathbb{R}^d \rightarrow \overline{\mathbb{R}}$ proper closed convex. Then if there exists $\bar{x} \in \text{dom}(f) \cap \text{dom}(g)$ with f continuous at $\bar{x}$ we have $\partial(f + g)(x) = \partial f(x) + \partial g(x)$.

Proof. The proof is not difficult but very technical and not short, uses Hahn-Banach, no further details. ■

Proof. (Monreau identity)

Fix $\hat{x} = \text{prox}_f(x)$, then

$$\begin{aligned}
 0 &\in \partial \left(f(\cdot) + \frac{1}{2} \|\cdot - x\|^2 \right) (\hat{x}) \\
 \implies 0 &\in \partial f(\hat{x}) + \frac{1}{2} \underbrace{(\hat{x} - x)}_{=\nabla \|\cdot\|^2} \\
 \implies x - \hat{x} &\in \partial f(\hat{x}).
 \end{aligned}$$

Recall that since f is proper convex, then $f = f^{**}$. Let $y = x - \hat{x}$. Then $y \in \partial f(\hat{x}) \iff \hat{x} \in \partial f^*(y)$, hence

$$\begin{aligned}
 x - y &\in \partial f^*(y) \\
 \implies 0 &\in \partial f^*(y) + (y - x) \\
 \implies 0 &\in \partial \left(f^*(\cdot) + \frac{1}{2} \|\cdot - x\|^2 \right) (y)
 \end{aligned}$$

hence $y = \text{prox}_{f^*}(x)$. In the end,

$$\text{prox}_f(x) + \text{prox}_{f^*}(x) = \hat{x} + y = \hat{x} + x - \hat{x} = x.$$

■

In general $x = \text{prox}_{\lambda f}(x) + \text{prox}_{\frac{1}{\lambda} f^*} \left(\frac{x}{\lambda} \right)$.

LECTURE 21 OF 12/05/2026

21.1 PROXIMAL ALGORITHMS

We know that x^* minimizes $f \iff x^*$ is a fixed point of $\text{prox}_{\lambda f}$.

Idea: x^0 given, iterate a fixed point iteration $x^{k+1} = \text{prox}_{\lambda f}(x^k)$. We know that this works if $\text{prox}_{\lambda f}$ is a contraction!

PROPOSITION 1. *The proximal mapping is 1-Lipschitz continuous (assuming f convex proper closed, $\lambda > 0$), i.e. $\|\text{prox}_{\lambda f}(x) - \text{prox}_{\lambda f}(y)\| \leq \|x - y\|$.*

Proof.

$$\begin{aligned}\hat{x} = \text{prox}_{\lambda f}(x) &\implies \langle x - \hat{x}, \hat{y} - \hat{x} \rangle + \lambda (f(\hat{x}) - f(\hat{y})) \leq 0 \\ \hat{y} = \text{prox}_{\lambda f}(y) &\implies \langle y - \hat{y}, \hat{x} - \hat{y} \rangle + \lambda (f(\hat{y}) - f(\hat{x})) \leq 0.\end{aligned}$$

Taking the sum of the two, we end up with the relation

$$\langle \hat{x} - x, \hat{x} - \hat{y} \rangle + \langle y - \hat{y}, \hat{x} - \hat{y} \rangle \leq 0.$$

Consider $\|\hat{x} - \hat{y}\|^2$, then

$$\begin{aligned}\|\hat{x} - \hat{y}\|^2 &= \langle \hat{x} - \hat{y}, \hat{x} - \hat{y} \rangle \quad \pm x, \pm y \text{ in the first term and expand} \\ &= \langle \hat{x} - x, \hat{x} - \hat{y} \rangle + \langle y - \hat{y}, \hat{x} - \hat{y} \rangle + \langle x - y, \hat{x} - \hat{y} \rangle \\ &\leq \langle x - y, \hat{x} - \hat{y} \rangle \leq \|x - y\| \cdot \|\hat{x} - \hat{y}\|.\end{aligned}$$

Therefore $\|\hat{x} - \hat{y}\| \leq \|x - y\|$. ■

21.1.1 Proximal point method

x^0 given, $x^{k+1} = \text{prox}_{t_k f}(x^k)$ for $\{t_k\}_k$ a given sequence of steps. Recall that $\nabla M_{\lambda f}(x) = \frac{1}{\lambda}(x - \text{prox}_{\lambda f}(x)) \implies x^{k+1} = x^k - t_k \nabla M_{t_k f}(x^k)$.

PROPOSITION 2. $f(x^{k+1}) \leq f(x^k) - \frac{1}{t_k} \|x^{k+1} - x^k\|^2$.

Proof. Immediate consequence of variational inequality. ■

PROPOSITION 3. Let f be proper closed convex, $f_{\text{opt}} = f(x^*) = \min_x f(x)$. Then it holds that

$$\begin{aligned} \|x^{k+1} - x^*\| &\leq \|x^k - x^*\| \\ f(x^{N+1}) - f_{\text{opt}} &\leq \frac{\|x^0 - x^*\|^2}{2 \sum_{k=1}^N t_k} \end{aligned}$$

where N is the total number of steps.

21.2 SPLITTING METHODS

Assume that we can decompose the objective function as $f(x) + g(x)$, where f is "simple in the prox sense", i.e. it is easy to compute $\text{prox}_{\lambda f}$ and g is smooth.

Our main assumption is $g : \mathbb{R}^d \rightarrow \mathbb{R}$ convex and differentiable. Moreover, we assume ∇g to be L -Lipschitz continuous, $L > 0$, i.e.

$$\|\nabla g(x) - \nabla g(y)\| \leq L \|x - y\|.$$

In this case we say that g is L -smooth.

LEMMA 1. If $g : \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth then

$$g(y) - g(x) - \langle \nabla g(x), y - x \rangle \leq \frac{L}{2} \|x - y\|^2.$$

Proof.

$$\begin{aligned} g(y) - g(x) - \langle \nabla g(x), y - x \rangle &= \int_0^1 \langle \nabla g(x + \tau(y - x)) - \nabla g(x), y - x \rangle d\tau \\ &= \int_0^1 \|\nabla g(x + \tau(y - x)) - \nabla g(x)\| \cdot \|y - x\| d\tau \\ &\leq L \int_0^1 \|\tau(y - x)\| \cdot \|y - x\| d\tau \\ &\leq \frac{L}{2} \|y - x\|^2 \end{aligned}$$
■

21.2 Splitting methods

Let now $\lambda > 0$, define $x^+ = x - \lambda \nabla g(x)$ a descend step for g . By the above lemma,

$$g(x^+) \leq g(x) + \langle \nabla g(x), x^+ - x \rangle + \frac{L}{2} \|x^+ - x\|^2 \quad (21.1)$$

$$\leq g(x) + \langle \nabla g(x), -\lambda \nabla g(x) \rangle + \frac{L}{2} \|\lambda \nabla g(x)\|^2 \quad (21.2)$$

$$= g(x) - \lambda \|\nabla g(x)\|^2 + \lambda^2 \frac{L}{2} \|\nabla g(x)\|^2 \quad (21.3)$$

$$= g(x) - \lambda \left(1 - \frac{L}{2} \lambda\right) \|\nabla g(x)\|^2 \quad (21.4)$$

which means that $g(x^+) \leq g(x)$ provided that $0 \leq \lambda \leq 2/L$. What about $f + g$? Substitute g with an upper bound,

$$g(x^k) + \langle \nabla g(x^k), x - x^k \rangle + \frac{1}{2\lambda} \|x - x^k\|^2$$

in the spirit of the above lemma.

Then

$$\begin{aligned} x^{k+1} &= \operatorname{argmin}_x \left[f(x) + g(x^k) + \langle \nabla g(x^k), x - x^k \rangle - \frac{1}{2\lambda} \|x - x^k\|^2 \right] \\ &= \operatorname{argmin}_x \left[f(x) + \langle \nabla g(x^k), x - x^k \rangle - \frac{1}{2\lambda} \|x - x^k\|^2 \right]. \end{aligned}$$

Observe that

$$\begin{aligned} \langle \lambda \nabla g(x^k), x - x^k \rangle &= \frac{1}{2} \|x - x^k\|^2 + 2 \langle \lambda \nabla g(x^k), x - x^k \rangle + \|\lambda \nabla g(x^k)\|^2 - \|\lambda \nabla g(x^k)\|^2 \\ &= \frac{1}{2} \|x - (x^k - \lambda \nabla g(x^k))\|^2 - \frac{1}{2} \lambda^2 \|\nabla g(x^k)\|^2. \end{aligned}$$

In the end, $x^{k+1} = \operatorname{argmin}_x \left[\lambda f(x) + \frac{1}{2} \|x - (x^k - \lambda \nabla g(x^k))\|^2 \right]$. Therefore,

$$x^{k+1} = \operatorname{prox}_{\lambda f}(x^k - \lambda \nabla g(x^k)).$$

This iteration is called proximal gradient method.

Example 2 (Regularized least squares). We are interested in

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|^2$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$. Define $g(x) = \frac{1}{2}\|Ax - b\|^2$, so that $\nabla g(x) = A^\top (Ax - b) \implies A^\top Ax = A^\top b$ is the minimum point.

The data b could be noisy or corrupted, so we add a regularizing term. The problem becomes

$$\min_x \frac{1}{2}\|Ax - b\|^2 + \alpha R(x)$$

where the first takes the role of $g(x)$ and the second $f(x)$.

21.3 PRIMAL-DUAL CHAMBOLLE-POCK ALGORITHM

We are interested in the problem

$$\min_x h(x) + K(Ax)$$

where h, K are convex proper closed functions, A a matrix. The idea is: instead of optimize directly, we look at the Lagrangian and we try to optimize for a saddle point.

So, given (x^k, y^k) optimize for $\max_y L(x^k, y)$. Observe that

$$\max_y h(x^k) - K^*(-y) - y^\top Ax^k = -\min_y K^*(y) - y^\top Ax^k.$$

Define $y^{k+1} \approx -\operatorname{argmin}_y K^*(y) - y^\top Ax^k$ as done via the proximal gradient step on y , therefore

$$y^{k+1} = -\operatorname{prox}_{\sigma K^*}(y^k + \sigma Ax^k).$$

Given then (x^k, y^{k+1}) we do a primal descend step on x , hence

$$x^{k+1} \approx \operatorname{argmin}_x L(x, y^{k+1}) \tag{21.5}$$

$$\approx h(x) - \underbrace{K^*(-y^{k+1})}_{\text{not depending on } x} - (y^{k+1})^\top Ax \tag{21.6}$$

$$\approx \operatorname{argmin}_x h(x) - (y^{k+1})^\top Ax \tag{21.7}$$

$$\implies x^{k+1} = \operatorname{prox}_{\tau h}(x^k + A^\top y^{k+1}). \tag{21.8}$$

The final algorithm then reads: choose $\tau, \sigma > 0$, $\vartheta \in [0, 1]$, (x^0, y^0) and set $\bar{x}^0 = x^0$. Then iterate

21.3 Primal-dual Chambolle-Pock algorithm

$$\begin{aligned}y^{k+1} &= \text{prox}_{\sigma K^*}(y^k + \sigma A \bar{x}^k) \\x^{k+1} &= \text{prox}_{\tau h}(x^k - \tau A^\top y^{k+1}) \\ \bar{x}^{k+1} &= x^{k+1} + \vartheta(x^{k+1} - x^k).\end{aligned}$$

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Do we have convergence? Just a few words: if we choose $\theta = 1$, set $L = \|A\|$ and it holds that $\tau\sigma L^2 < 1$, then $(x^k, y^k) \rightarrow (x^*, y^*)$ as C/N (order 1 convergence). The proof is quite long and not very interesting.

Example 1 (ROF model, image denoising). We can consider an image (black and white) as a map $g : \Omega \rightarrow [0, 1]$. Imagine that in the acquisition process for g an arbitrary amount of noise is added. We can recover the "true" image as the minimizer of the functional

$$\frac{\lambda}{2} \int_{\Omega} |u - g|^2 \, dx + \int_{\Omega} |Du|$$

where $|Du|$ denotes the total variation measure. In general we seek for a minimizer $u \in \text{BV}(\Omega)$, but, if $u \in W^{1,1}(\Omega)$, then $|Du| = \|\nabla u\| \, dx$

We consider first the function u to be a $W^{1,1}$ function, then, WLOG, $\Omega = [0, 1] \times [0, 1]$. We discretize Ω with a uniform grid in each dimension, so that $\Omega \approx (x_i, y_j)_{ij} = (ih, jh)_{ij}$ where h is the discretization step. Then we define the function on the grid as a matrix of samples $u = \{u_{ij}\}_{ij}$ and the noisy function $g = \{g_{ij}\}_{ij}$. The problem then becomes

$$\frac{\lambda}{2} \sum_{ij} h^2 |u_{ij} - g_{ij}| + \sum_{ij} h^2 \|(\nabla u)_{ij}\|.$$

Let $X = \mathbb{R}^{M \times N}$, $Y = (\mathbb{R}^{M \times N}, \mathbb{R}^{M \times N})$, $p \in Y \implies p = (p^1, p^2)$, and define the map $A : X \rightarrow Y$, where $Au = (\nabla u)_{ij}$ with forward finite differences, so

$$(\nabla u)_{ij}^1 = \begin{cases} \frac{u_{i+1,j} - u_{ij}}{h} & \text{if } i < M \\ 0 & \text{otherwise, Neumann BC} \end{cases}$$

and

$$(\nabla u)_{ij}^2 = \begin{cases} \frac{u_{i,j+1} - u_{i,j}}{h} & \text{if } j < N \\ 0 & \text{otherwise, Neumann BC} \end{cases}$$

In order to apply the Chambolle-Pock algorithm, it is needed to compute the discrete adjoint operator ∇^* , which could be very large to memorize in practice. Even for a 1024×1024 pixels image, it would require approximately 8 terabytes of memory to store the full dense matrix. Even when considering a sparse matrix representation, storing the explicit structure results in a significant waste of memory.

Instead, the adjoint operator can be implemented in a *matrix-free* fashion by exploiting its analytical definition in the discrete setting. By leveraging the definition of the adjoint via the standard inner product:

$$\langle \nabla u, p \rangle = \langle u, \nabla^* p \rangle \quad (22.1)$$

it can be shown that the transpose of the Forward Finite Difference (FFD) matrix used for the gradient corresponds exactly to the negative divergence operator calculated via Backward Finite Differences (BFD).

Therefore, rather than storing a massive matrix and performing costly matrix-vector multiplications, ∇^* can be computed on the fly using a highly efficient, pixel-by-pixel local stencil operation. This approach reduces the operator's memory footprint to zero, significantly optimizes cache reuse, and makes the algorithm ideally suited for massive parallelization on modern GPU architectures.

Therefore we seek for the adjoint operator. Notice that

$$\begin{aligned} \langle u, v \rangle_X &= \sum_{ij} u_{ij} v_{ij} \\ \langle p, q \rangle_Y &= \sum_{ij} p_{ij}^1 q_{ij}^1 + p_{ij}^2 q_{ij}^2. \end{aligned}$$

The goal is to have something in the form $\sum_{ij} u_{ij} (\nabla^* p)_{ij}$. Idea: take only the terms where u_{ij} appears, i.e. $-p_{ij}^1, -p_{ij}^2$ and $p_{i-1,j}^1, p_{i,j-1}^2$ as when i consider the $i-1$ i get $u_{ij} - u_{i-1,j}$, therefore a term with index (ij) appears. Therefore

$$(\nabla^* p)_{ij} = - \left(\frac{p_{ij}^1 - p_{i-1,j}^1}{h} \right) - \left(\frac{p_{ij}^2 - p_{i,j-1}^2}{h} \right)$$

which is the negative discrete divergence computed with backward finite differences.